



Advanced Numerical Analysis

Introduction to FEM

Author: Ziyi Wang

Institute: UESTC SMS

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Bio: Information



电子科技大学 数学科学学院
School of Mathematical Sciences UESTC

Stay hungry, stay foolish.

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Chapter 1 Ritz-Galerkin Methods for Second-Order ODEs in 1D

1.1 Model Problem and Weak Formulation

1.1.1 The Boundary Value Problem

A two-point Boundary Value Problem (BVP) is often used to describe various physical models, such as:

- The deformation of an elastic bar under load.
- Heat conduction in a rod.
- The displacement of an elastic cord.

Let us consider the following model problem for an elastic cord:

$$\begin{cases} -u''(x) = f(x), & x \in (0, 1), \\ u(0) = u(1) = 0. \end{cases} \quad (1.1)$$

We refer to (1.1) as the **strong form** of the problem.

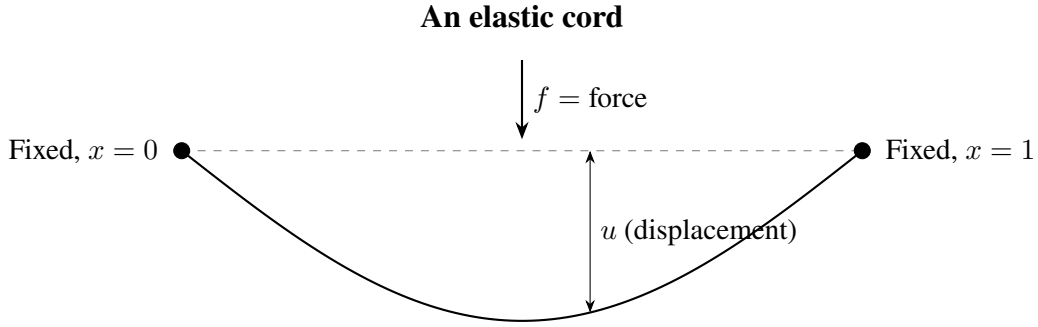


Figure 1.1: Deformation of a Fixed-Fixed Elastic Cord

1.1.2 Derivation of the Weak Formulation

Suppose u is the classical solution and v is any smooth “test function” satisfying the boundary conditions $v(0) = v(1) = 0$. Multiplying the equation by v and integrating over $(0, 1)$, integration by parts yields:

$$\begin{aligned} (f, v) &:= \int_0^1 f(x)v(x) \, dx \\ &= \int_0^1 -u''(x)v(x) \, dx \\ &= [-u'(x)v(x)]_0^1 + \int_0^1 u'(x)v'(x) \, dx \\ &= \int_0^1 u'(x)v'(x) \, dx =: a(u, v), \end{aligned}$$

where the boundary term vanishes because $v(0) = v(1) = 0$.

Let us formally define the “energy space” (mathematically known as the Sobolev space $H_0^1(0, 1)$):

$$V = \{v \in L^2(0, 1) : a(v, v) < +\infty \text{ and } v(0) = v(1) = 0\}.$$

Here, $v \in L^2(0, 1)$ means v is Lebesgue measurable and square-integrable, i.e., $\int_0^1 |v(x)|^2 \, dx < +\infty$.

Remark We shall formally introduce notations like Sobolev spaces in future lectures. For now, V can be thought of as the space of functions with square-integrable first derivatives satisfying the boundary conditions.

The **weak formulation** (or variational formulation) is stated as follows:

$$\text{Find } u \in V \text{ such that } a(u, v) = (f, v) \quad \text{for all } v \in V. \quad (1.2)$$

A solution u to (1.2) is called a *weak solution*.

Remark If u is a strong solution to (1.1), it implies that u is a solution to (1.2). However, the converse is not necessarily true (pointwise). A weak solution requires less regularity.

Example: Consider the problem:

$$\begin{cases} -u''(x) = f(x) = \frac{1}{\sqrt{|x|}}, & x \in (-1, 1), \\ u(-1) = u(1) = 0. \end{cases}$$

We can verify that $u(x) = \frac{4}{3}(1 - |x|^{3/2})$ is a weak solution. However, it is **not** a strong solution in the classical sense because $u''(x)$ blows up at $x = 0$.

1.1.3 Regularity and Variational Principle

If the data f is sufficiently smooth, the weak solution recovers the strong solution.

Theorem 1.1 (Regularity)

Suppose $f \in C^0([0, 1])$ and $u \in C^2([0, 1])$ satisfy Problem (1.2). Then u solves the strong Problem (1.1).



Proof Choose any $v \in V \cap C^0([0, 1])$. Integration by parts on the weak form gives:

$$(f, v) = a(u, v) = \int_0^1 (-u'')v \, dx.$$

Thus, rearranging terms:

$$\int_0^1 (f + u'')v \, dx = 0, \quad \forall v \in V \cap C^0([0, 1]). \quad (1.3)$$

Since $f + u''$ is continuous on $[0, 1]$, suppose for the sake of contradiction that there exists x_0 such that $(f + u'')(x_0) \neq 0$. By continuity, there exists an interval (x_1, x_2) where $f + u''$ maintains the same sign.

Construct a bump function $v \in V$:

$$v(x) = \begin{cases} (x - x_1)^2(x - x_2)^2, & \text{if } x \in (x_1, x_2), \\ 0, & \text{otherwise.} \end{cases}$$

Then, the integral $\int_{x_1}^{x_2} (f + u'')v \, dx$ is strictly non-zero, contradicting (1.3). Hence, $f(x) + u''(x) = 0$ for all $x \in (0, 1)$.

Why prefer the weak formulation?

1. It requires less regularity (derivatives) of the solution.
2. It is the basis for the Finite Element Method.
3. It is equivalent to a minimization problem (principle of minimum potential energy).

Theorem 1.2 (Variational Principle)

Define the energy functional $F : V \rightarrow \mathbb{R}$ as

$$F(v) = \frac{1}{2}a(v, v) - (f, v).$$

Consider the minimization problem:

$$\text{Find } u \in V \quad \text{s.t.} \quad F(u) \leq F(v) \quad \text{for all } v \in V. \quad (1.4)$$

Then, Problem (1.2) is equivalent to Problem (1.4). 

Proof ($\Rightarrow$) Let u solve (1.2). Let $w = v - u \in V$. Then:

$$\begin{aligned} F(v) &= F(u + w) = \frac{1}{2}a(u + w, u + w) - (f, u + w) \\ &= \frac{1}{2}a(u, u) + a(u, w) + \frac{1}{2}a(w, w) - (f, u) - (f, w) \\ &= \underbrace{\left[\frac{1}{2}a(u, u) - (f, u) \right]}_{F(u)} + \underbrace{[a(u, w) - (f, w)]}_{=0} + \underbrace{\frac{1}{2}a(w, w)}_{\geq 0} \\ &\geq F(u). \end{aligned}$$

($\Leftarrow$) Let u be the minimizer of (1.4). For any $v \in V$ and $\varepsilon \in \mathbb{R}$, define $g(\varepsilon) = F(u + \varepsilon v)$. Since u is a minimizer, $g(\varepsilon)$ has a minimum at $\varepsilon = 0$, implying $g'(0) = 0$.

$$\begin{aligned} g(\varepsilon) &= \frac{1}{2}a(u + \varepsilon v, u + \varepsilon v) - (f, u + \varepsilon v) \\ &= \frac{1}{2}a(u, u) + \varepsilon a(u, v) + \frac{1}{2}\varepsilon^2 a(v, v) - (f, u) - \varepsilon(f, v). \end{aligned}$$

The derivative is $g'(\varepsilon) = a(u, v) + \varepsilon a(v, v) - (f, v)$. Setting $g'(0) = 0$ yields:

$$a(u, v) - (f, v) = 0 \quad \implies \quad a(u, v) = (f, v).$$

Remark Problem (1.4) corresponds to the “Principle of Minimum Potential Energy in Mechanics.”

 **Exercise 1.1** Prove that Problem (1.2) has a **unique** solution. (Hint: Use the Lax-Milgram Theorem or strictly convexity of F).

1.2 The Ritz-Galerkin Approximation

Let us replace the infinite-dimensional energy space V with a finite-dimensional subspace $S \subseteq V$. The discrete problem reads:

$$\text{Find } u_S \in S \text{ such that } a(u_S, v) = (f, v) \quad \text{for all } v \in S. \quad (1.5)$$

Here, u_S is the **Ritz-Galerkin approximation** of u .

Let $\mathcal{B} = \{\phi_j\}_{j=1}^n$ be a basis of S . We can expand the solution and test function as:

$$u_S = \sum_{j=1}^n u_j \phi_j, \quad v = \sum_{i=1}^n v_i \phi_i,$$

where $\{u_j\}$ are the unknown coefficients to be determined. Substituting these into (1.5) and testing with each basis function ϕ_i yields the linear system:

$$\sum_{j=1}^n u_j a(\phi_j, \phi_i) = (f, \phi_i), \quad i = 1, \dots, n.$$

In matrix form:

$$AU = F,$$

where

- $U = (u_1, \dots, u_n)^\top$ is the coefficient vector.
- $A \in \mathbb{R}^{n \times n}$ is the **Stiffness Matrix** with entries $A_{ij} = a(\phi_j, \phi_i)$.
- $F \in \mathbb{R}^n$ is the load vector with entries $F_i = (f, \phi_i)$.

Proposition 1.1

The linear system $AU = F$ admits a unique solution.

Proof The matrix A is symmetric by the symmetry of $a(\cdot, \cdot)$. To show uniqueness, assume $F = 0$ (homogeneous system). This implies:

$$a(u_S, v) = 0, \quad \forall v \in S.$$

Choosing $v = u_S$, we get $a(u_S, u_S) = \int_0^1 (u'_S)^2 dx = 0$. This implies $u'_S = 0$ almost everywhere, so u_S is constant. Given the boundary conditions $u_S(0) = u_S(1) = 0$, we must have $u_S \equiv 0$. Thus, $U = 0$, and A is non-singular.

1.3 The Finite Element Method in 1D

In this course, we focus on a specific class of subspaces S constructed using subdivisions of the domain. This approach is called the **Finite Element Method (FEM)**.

Let $0 = x_0 < x_1 < \dots < x_n = 1$ be a partition (mesh) of $[0, 1]$. Define the mesh size $h_i = x_i - x_{i-1}$ and $h = \max_i h_i$. Let S_h be the space of continuous, piecewise linear functions subordinate to this partition:

$$S_h = \{v \in C^0([0, 1]) : v|_{[x_{i-1}, x_i]} \in \mathbb{P}^1, \forall i, \text{ and } v(0) = v(1) = 0\}.$$

Exercise: Verify that S_h is a subspace of V .

We choose the **nodal basis** (hat functions) $\{\phi_i\}_{i=1}^{n-1}$ defined by:

$$\phi_i(x_j) = \delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

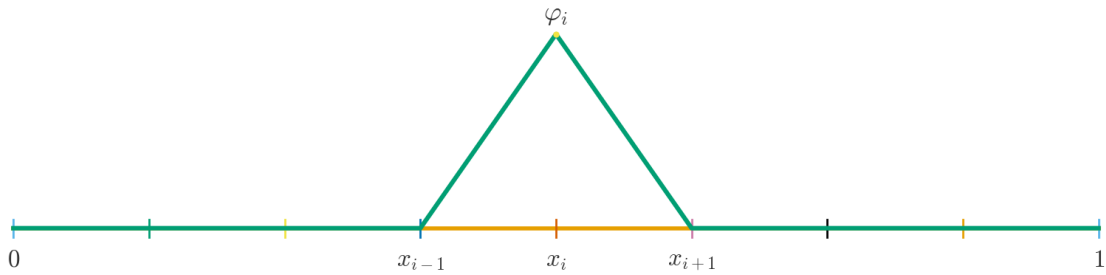


Figure 1.2: Illustration of a 1D linear basis function (Hat function).

Definition 1.1

Define the **Interpolation Operator** $\mathcal{I} : C^0([0, 1]) \rightarrow S$ as:

$$\mathcal{I}v(x) = \sum_{i=1}^{n-1} v(x_i) \phi_i(x).$$

Note that $\mathcal{I}v(x_i) = v(x_i)$ at the nodes.



1.4 A Priori Error Analysis

We now analyze the error between the exact solution u and the discrete solution u_S .

1.4.1 Interpolation Error

Assume $v \in C^2([0, 1])$. The error of the linear interpolant is bounded as follows. On an interval $[x_{i-1}, x_i]$, the derivative error is:

$$(\mathcal{I}_h v)'(x) - v'(x) = v''(\eta)(x - \xi),$$

where $\xi \in (x_{i-1}, x_i)$ is a point where the derivatives match (by Mean Value Theorem), and η is an intermediate point. Integrating the square of the error:

$$\int_{x_{i-1}}^{x_i} |(\mathcal{I}_h v)' - v'|^2 dx \leq h^2 \int_{x_{i-1}}^{x_i} |v''|^2 dx.$$

Summing over all elements, we obtain the estimate in the energy norm $\|v\|_E := (\int_0^1 (v')^2 dx)^{1/2}$:

$$\|v - \mathcal{I}_h v\|_E \leq Ch \left(\int_0^1 |v''|^2 dx \right)^{1/2}.$$

1.4.2 Cea's Lemma (Energy Norm Estimate)

Now we are ready to show the approximation error between u and u_S . Let us write the two problem again:

$$a(u, v) = (f, v) \quad \forall v \in V.$$

$$a(u_S, v) = (f, v) \quad \forall v \in S_h.$$

Since $S_h \subset V$, we set $v \in S_h \subset V$ and subtract the above two equations, we have the **Galerkin Orthogonality**:

$$a(u - u_S, v) = 0, \quad \forall v \in S_h.$$

Using this, we can derive the best approximation property (Cea's Lemma):

$$\begin{aligned} \|u - u_S\|_E^2 &= a(u - u_S, u - u_S) \\ &= a(u - u_S, u - v) + \underbrace{a(u - u_S, v - u_S)}_{=0} \\ &\leq \|u - u_S\|_E \|u - v\|_E. \end{aligned}$$

Thus, $\|u - u_S\|_E \leq \inf_{v \in S_h} \|u - v\|_E$. Choosing $v = \mathcal{I}_h u$:

$$\|u - u_S\|_E \leq Ch \|u''\|_{L^2}.$$

1.4.3 The Duality Argument (L^2 Norm Estimate)

To estimate $\|u - u_S\|_{L^2}$, we use Nitsche's duality trick. Consider the dual problem: find $z \in V$ such that

$$-z'' = u - u_S \quad \text{in } (0, 1), \quad z(0) = z(1) = 0.$$

This implies $a(v, z) = (u - u_S, v)$ for all $v \in V$. Regularity theory gives $\|z''\|_{L^2} \leq \|u - u_S\|_{L^2}$.

Testing with $v = u - u_S$:

$$\begin{aligned} \|u - u_S\|_{L^2}^2 &= (u - u_S, u - u_S) = a(u - u_S, z) \\ &= a(u - u_S, z - \mathcal{I}_h z) \quad (\text{by orthogonality}) \\ &\leq \|u - u_S\|_E \|z - \mathcal{I}_h z\|_E \\ &\leq (Ch \|u''\|_{L^2}) \cdot (Ch \|z''\|_{L^2}) \\ &\leq Ch^2 \|u''\|_{L^2} \|u - u_S\|_{L^2}. \end{aligned}$$

Dividing by $\|u - u_S\|_{L^2}$ gives the quadratic convergence rate:

Theorem 1.3

If $u \in H^2(0, 1)$, then:

$$\|u - u_S\|_{L^2} + h \|u - u_S\|_E \leq Ch^2 \|u''\|_{L^2}.$$



Remark (other finite dimensional subspace S_h)

1. $\phi_j(x) = \sin(j\pi x)$, $j = 1, \dots, n$, $x \in (0, 1)$.
2. $\phi_j(x) = x^j(1 - x)$, $j = 1, \dots, n$, $x \in (0, 1)$.

1.5 Implementation and Comparison

1.5.1 Stiffness Matrix Assembly

Let us try to compute the discrete system using the finite element space. Note that

$$A_{ij} = a \int_0^1 \phi_j' \phi_i' dx \quad F_i = \int_0^1 f \phi_i dx.$$

for $i, j = 1, \dots, n - 1$.

The shape functions are

$$\phi_j(x) = \begin{cases} \frac{x - x_{j-1}}{x_j - x_{j-1}} & x \in [x_{j-1}, x_j] \\ \frac{x_{j+1} - x}{x_{j+1} - x_j} & x \in [x_j, x_{j+1}] \\ 0, & \text{otherwise.} \end{cases}$$

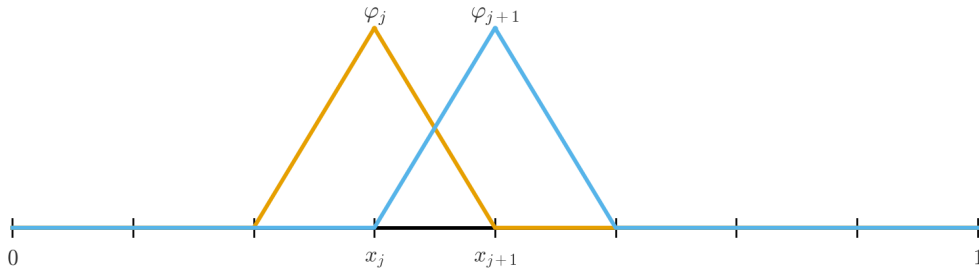


Figure 1.3: Basis functions for the linear hat functions.

For the linear hat functions, the derivative is piecewise constant:

$$\phi'_j(x) = \begin{cases} \frac{1}{h_j}, & x \in [x_{j-1}, x_j] \\ -\frac{1}{h_{j+1}}, & x \in [x_j, x_{j+1}]. \end{cases}$$

What's more, if $|i - j| > 1$, then $\int_0^1 \phi'_i \phi'_j dx = 0$. The stiffness matrix entries are calculated as:

$$A_{jj} = \int_0^1 (\phi'_j)^2 dx = \frac{1}{h_j} + \frac{1}{h_{j+1}},$$

$$A_{j,j+1} = A_{j+1,j} = \int_0^1 \phi'_j \phi'_{j+1} dx = -\frac{1}{h_{j+1}}.$$

The Matrix form is as follows:

$$A = \begin{pmatrix} \frac{1}{h_1} + \frac{1}{h_2} & -\frac{1}{h_2} & & \\ -\frac{1}{h_2} & \frac{1}{h_2} + \frac{1}{h_3} & -\frac{1}{h_3} & \\ & \ddots & \ddots & \\ & & -\frac{1}{h_n} & \frac{1}{h_{n-1}} + \frac{1}{h_n} \end{pmatrix}$$

The following properties hold:

1. A is symmetric.
2. A is a tri-diagonal matrix.
3. A is positive definite $\Rightarrow A$ is non-singular.

For a uniform mesh with size h :

$$A = \frac{1}{h} \begin{pmatrix} 2 & -1 & 0 & \cdots \\ -1 & 2 & -1 & \cdots \\ 0 & -1 & 2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

1.5.2 Comparison with Finite Difference Method

The standard central difference approximation for $-u'' = f$ is:

$$-\frac{u_{j+1} - 2u_j + u_{j-1}}{h^2} = f(x_j).$$

In the FEM formulation, the j -th row of the system reads:

$$\frac{1}{h}(-u_{j-1} + 2u_j - u_{j+1}) = (f, \phi_j).$$

If we approximate the load integral using the trapezoidal rule, $(f, \phi_j) \approx hf(x_j)$, the FEM system becomes identical to the Finite Difference system.

Chapter 2 Sobolev Spaces

2.1 Sobolev Spaces

Let $\Omega \subset \mathbb{R}^n$ ($n \geq 1$) be a bounded open domain.

2.1.1 $L^p(\Omega)$ spaces

Definition 2.1

A (real-valued) measurable function $f : \Omega \rightarrow \mathbb{R}$ belongs to $L^p(\Omega)$ if

$$\int_{\Omega} |f(x)|^p dx < \infty \quad (1 \leq p < \infty).$$

For $p = \infty$ we define

$$L^\infty(\Omega) = \left\{ f : \Omega \rightarrow \mathbb{R} \text{ measurable} : \operatorname{ess\,sup}_{x \in \Omega} |f(x)| < \infty \right\}.$$



For $1 \leq p < \infty$ we set

$$\|f\|_{L^p(\Omega)} = \left(\int_{\Omega} |f(x)|^p dx \right)^{1/p},$$

and for $p = \infty$ we set

$$\|f\|_{L^\infty(\Omega)} = \operatorname{ess\,sup}_{x \in \Omega} |f(x)|.$$

If $\overline{\Omega}$ is compact and $f \in C(\overline{\Omega})$, then

$$\|f\|_{L^\infty(\Omega)} = \max_{x \in \overline{\Omega}} |f(x)|.$$

Theorem 2.1 (Minkowski inequality)

Let $1 \leq p \leq \infty$ and $f, g \in L^p(\Omega)$. Then

$$\|f + g\|_{L^p(\Omega)} \leq \|f\|_{L^p(\Omega)} + \|g\|_{L^p(\Omega)}.$$

More generally, for $f_k \in L^p(\Omega)$, $k = 1, \dots, N$,

$$\left\| \sum_{k=1}^N f_k \right\|_{L^p(\Omega)} \leq \sum_{k=1}^N \|f_k\|_{L^p(\Omega)}.$$



Remark If $\int_{\Omega} |f|^p dx = 0$, then $f = 0$ almost everywhere in Ω . Hence $\|\cdot\|_{L^p(\Omega)}$ is indeed a norm on $L^p(\Omega)$.

Theorem 2.2 (Hölder inequality)

Let $1 < p, q < \infty$ with $1/p + 1/q = 1$ and $f \in L^p(\Omega)$, $g \in L^q(\Omega)$. Then

$$\int_{\Omega} |f(x)g(x)| dx \leq \|f\|_{L^p(\Omega)} \|g\|_{L^q(\Omega)}.$$



For $p = q = 2$, Hölder inequality becomes the Cauchy–Schwarz inequality

$$\left| \int_{\Omega} f(x)g(x) dx \right| \leq \|f\|_{L^2(\Omega)} \|g\|_{L^2(\Omega)}.$$

Example 2.1

$$\begin{aligned} \left(\int_{\Omega} f(x) \cdot dx \right)^2 &\leq \left(\int_{\Omega} |1 \cdot f(x)| dx \right)^2 \\ &\leq \int_{\Omega} 1 dx \int_{\Omega} |f(x)|^2 dx \leq \mu(\Omega) \int_{\Omega} |f|^2 dx \end{aligned}$$

Definition 2.2

For $f, g \in L^2(\Omega)$ we define

$$(f, g)_{L^2(\Omega)} := \int_{\Omega} f(x)g(x) dx.$$

Then $(\cdot, \cdot)_{L^2(\Omega)}$ is an inner product on $L^2(\Omega)$.

**Theorem 2.3**

For $1 \leq p \leq \infty$, $L^p(\Omega)$ is complete with respect to the norm $\|\cdot\|_{L^p(\Omega)}$, i.e. $L^p(\Omega)$ is a Banach space. namely every Cauchy sequence (w.r.t. $\|\cdot\|_{L^p}$) in $L^p(\Omega)$, say $\{f_n\} \subset L^p(\Omega)$, converges to a function (unique) $f \in L^p(\Omega)$, i.e.

$$\lim_{n \rightarrow \infty} \|f - f_n\|_{L^p} = 0$$

For $p = 2$, $L^2(\Omega)$ equipped with the inner product $(\cdot, \cdot)_{L^2(\Omega)}$ is a Hilbert space.



We denote the Lebesgue measure of a measurable set $E \subset \Omega$ by $\mu(E)$.

2.1.2 Generalized or weak derivatives

Let $\Omega \subset \mathbb{R}^d$ and let $\alpha = (\alpha_1, \dots, \alpha_d)$ be a multi-index with $|\alpha| = \alpha_1 + \dots + \alpha_d$.

Definition 2.3

For a sufficiently smooth function u we define

$$D^{\alpha}u = \frac{\partial^{|\alpha|}u}{\partial x_1^{\alpha_1} \dots \partial x_d^{\alpha_d}},$$

and write $D^0u = u$.

**Definition 2.4**

We denote by $C_c^{\infty}(\Omega)$ the space of infinitely differentiable functions $\varphi : \mathbb{R}^d \rightarrow \mathbb{R}$ whose support is contained in Ω .

**Definition 2.5**

For $\varphi \in C_c^{\infty}(\Omega)$ the support of φ is defined by

$$\text{supp}(\varphi) := \overline{\{x \in \Omega : \varphi(x) \neq 0\}} \subset \overline{\Omega}.$$

**Definition 2.6 (L_{loc}^1)**

A measurable function f is said to belong to $L_{\text{loc}}^1(\Omega)$ if

$$\int_K |f(x)| dx < \infty \quad \text{for every compact set } K \subset \Omega.$$



Example 2.2 The function $f(x) = \frac{1}{x}$ on $\Omega = (0, 1)$ satisfies $f \in L^1_{\text{loc}}(0, 1)$ but $f \notin L^1(0, 1)$ since

$$\int_a^1 \frac{1}{x} dx < \infty \quad \text{for all } a > 0, \quad \int_0^1 \frac{1}{x} dx = \infty.$$

 **Exercise 2.1** For what value of α Satisfying $x^{-\alpha} \in L^p(0, 1)$?

Definition 2.7 (Weak derivative)

Let $f \in L^1_{\text{loc}}(\Omega)$ and let α be a multi-index. A function $g \in L^1_{\text{loc}}(\Omega)$ is called the weak derivative of f of order α , written $g = D^\alpha f$, if

$$\int_{\Omega} f(x) D^\alpha \varphi(x) dx = (-1)^{|\alpha|} \int_{\Omega} g(x) \varphi(x) dx \quad \text{for all } \varphi \in C_c^\infty(\Omega).$$



Examples in one dimension

Let $\Omega = \mathbb{R}$ for simplicity.

Example 2.3 Smooth case If $f \in C^1(\mathbb{R})$ with compact support, then integration by parts shows

$$\int_{\mathbb{R}} f(x) \varphi'(x) dx = - \int_{\mathbb{R}} f'(x) \varphi(x) dx \quad \forall \varphi \in C_c^\infty(\mathbb{R}),$$

so the usual derivative f' is also the weak derivative.

Example 2.4A continuous, piecewise smooth function Define

$$g(x) = \begin{cases} 0, & x \leq 0, \\ x, & 0 < x < 1, \\ 1, & x \geq 1. \end{cases}$$

Then g is continuous and piecewise C^1 . A direct computation using integration by parts on $(0, 1)$ shows that

$$\int_{\mathbb{R}} g(x) \varphi'(x) dx = - \int_0^1 \varphi(x) dx \quad \forall \varphi \in C_c^\infty(\mathbb{R}).$$

Hence the weak derivative of g is

$$g'(x) = \chi_{(0,1)}(x),$$

the characteristic function of the open interval $(0, 1)$.

Example 2.5 Absolute value Let $f(x) = |x|$ on $\mathbb{R}$. For $\varphi \in C_c^\infty(\mathbb{R})$ we have

$$\int_{\mathbb{R}} |x| \varphi'(x) dx = \int_0^\infty x \varphi'(x) dx - \int_{-\infty}^0 x \varphi'(x) dx = - \int_0^\infty \varphi(x) dx + \int_{-\infty}^0 \varphi(x) dx = - \int_{\mathbb{R}} \text{sgn}(x) \varphi(x) dx.$$

Thus the weak derivative of $|x|$ is

$$f'(x) = \text{sgn}(x) = \begin{cases} -1, & x < 0, \\ 0, & x = 0, \\ 1, & x > 0, \end{cases}$$

which belongs to $L^1_{\text{loc}}(\mathbb{R})$ although the classical derivative does not exist at $x = 0$.

Example 2.6A step function without weak derivative Let

$$h(x) = \begin{cases} 0, & x < 0, \\ 1, & x \geq 0. \end{cases}$$

Then for $\varphi \in C_c^\infty(\mathbb{R})$,

$$\int_{\mathbb{R}} h(x) \varphi'(x) dx = -\varphi(0).$$

If a weak derivative $g \in L_{\text{loc}}^1(\mathbb{R})$ of h existed, we would have

$$-\varphi(0) = \int_{\mathbb{R}} h(x) \varphi'(x) dx = - \int_{\mathbb{R}} g(x) \varphi(x) dx \quad \forall \varphi \in C_c^\infty(\mathbb{R}).$$

Using a sequence of mollifiers $\{\varphi_\varepsilon\}$ that approximate the Dirac mass at 0 and the dominated convergence theorem leads to a contradiction. Hence h has no weak derivative in L_{loc}^1 .

2.1.3 Sobolev spaces $W^{k,p}(\Omega)$

Definition 2.8

Let $k \in \mathbb{N}$ and $1 \leq p \leq \infty$. The Sobolev space $W^{k,p}(\Omega)$ is defined by

$$W^{k,p}(\Omega) = \left\{ u \in L^p(\Omega) : D^\alpha u \in L^p(\Omega) \text{ for all multi-indices } \alpha \text{ with } |\alpha| \leq k \right\}.$$



If $d = 1$, and $k = 1$, $\Omega = (a, b)$, $W^{1,p}(\Omega) = \{f \in L^p(\Omega) \mid f' \text{ exists and } f' \in L^p(\Omega)\}$.

Remark If $u \in C^k(\overline{\Omega})$, then the weak derivatives $D^\alpha u$ coincide with the classical derivatives; hence $C^k(\overline{\Omega}) \subset W^{k,p}(\Omega)$.

Definition 2.9 (Norm and seminorm)

For $1 \leq p < \infty$ we define

$$\|u\|_{W^{k,p}(\Omega)} := \left(\sum_{|\alpha| \leq k} \|D^\alpha u\|_{L^p(\Omega)}^p \right)^{1/p}, \quad |u|_{W^{k,p}(\Omega)} := \left(\sum_{|\alpha|=k} \|D^\alpha u\|_{L^p(\Omega)}^p \right)^{1/p}.$$

For $p = \infty$ we use

$$\|u\|_{W^{k,\infty}(\Omega)} := \max_{|\alpha| \leq k} \|D^\alpha u\|_{L^\infty(\Omega)}, \quad |u|_{W^{k,\infty}(\Omega)} := \max_{|\alpha|=k} \|D^\alpha u\|_{L^\infty(\Omega)}.$$



Theorem 2.4

For each $k \in \mathbb{N}$ and $1 \leq p \leq \infty$, the space $W^{k,p}(\Omega)$ equipped with $\|\cdot\|_{W^{k,p}(\Omega)}$ is a Banach space.



For $p = 2$ we write

$$H^k(\Omega) := W^{k,2}(\Omega).$$

Definition 2.10

For $u, v \in H^k(\Omega)$ we define the inner product

$$(u, v)_{H^k(\Omega)} := \sum_{|\alpha| \leq k} (D^\alpha u, D^\alpha v)_{L^2(\Omega)}.$$

Then $H^k(\Omega)$ is a Hilbert space with this inner product.



Example 2.7 $f \in H^1(\Omega) \Rightarrow f \in L^2(\Omega)$ and $\frac{\partial}{\partial x_i} f \in L^2(\Omega), i = 1, 2, \dots, d$.

$$\begin{aligned} \|f\|_{H^1(\Omega)} &= (\|f\|_{L^2}^2 + \|\nabla f\|_{L^2}^2)^{1/2} \\ \|\nabla f\|_{L^2}^2 &= \int_{\Omega} |\nabla f|^2 dx \end{aligned}$$

2.1.4 A one-dimensional example and hat functions

Let $\Omega = (a, b) \subset \mathbb{R}$ and choose a partition

$$a = x_0 < x_1 < \cdots < x_N = b.$$

Define

$$\mathcal{M} = \left\{ v \in C([a, b]) : v|_{(x_{i-1}, x_i)} \text{ is linear for } i = 1, \dots, N \right\}.$$

For each node x_j ($j = 0, \dots, N$) we define the *hat function* $\varphi_j \in \mathcal{M}$ by

$$\varphi_j(x_i) = \delta_{ij}, \quad i = 0, \dots, N,$$

and on each interval (x_{i-1}, x_i) we let φ_j be the unique linear function interpolating these node values. Explicitly, for $j = 1, \dots, N-1$,

$$\varphi_j(x) = \begin{cases} \frac{x - x_{j-1}}{x_j - x_{j-1}}, & x \in [x_{j-1}, x_j], \\ \frac{x_{j+1} - x}{x_{j+1} - x_j}, & x \in [x_j, x_{j+1}], \\ 0, & \text{otherwise,} \end{cases}$$

and similar formulas hold for φ_0 and φ_N .

The derivatives φ_j' exist almost everywhere and are piecewise constant, hence $\varphi_j \in H^1(a, b)$ for all j . One can show that $\{\varphi_j\}_{j=0}^N$ is a basis of $\mathcal{M}$.

 **Exercise 2.2** The function ϕ_j has a weak derivative which equals $\phi_j'(x)$ for $x \in (x_l, x_{l+1})$, $l = 0, 1, \dots, n-1$. $\phi_j \in H^1(a, b)$.

Remark In fact $\phi_j \in W^{1,p}(a, b)$, $p \in [1, +\infty]$. $\Rightarrow \mathcal{M} \subset W^{1,p}(\Omega)$, $p \in [1, +\infty]$.

2.1.5 Domains with continuous / Lipschitz boundary

Let $\Omega \subset \mathbb{R}^2$ be a bounded domain and $x^0 \in \partial\Omega$.

Definition 2.11 (Continuous boundary)

We say that $\partial\Omega$ is continuous (of class C^0) if for each $x^0 \in \partial\Omega$ there exist a rectangle $V \subset \mathbb{R}^2$ containing x^0 , a new Cartesian coordinate system (x_1, x_2) with origin at x^0 , and a continuous function $\psi : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\Omega \cap V = \{y \in V : y_n > \psi(y')\}, \quad \partial\Omega \cap V = \{y \in V : y_n = \psi(y')\}.$$



Definition 2.12 (Lipschitz boundary)

We say that $\partial\Omega$ is Lipschitz if in the above definition the function ψ can be chosen to be Lipschitz continuous: there exists $L > 0$ such that

$$|\psi(y'_1) - \psi(y'_2)| \leq L |y'_1 - y'_2| \quad \text{for all } y'_1, y'_2.$$



A typical example of a domain without continuous boundary is a “slit domain”, e.g. a square with a line segment removed from its interior.

Theorem 2.5 (Density of smooth functions)

If Ω has a Lipschitz boundary, then $C^\infty(\overline{\Omega})$ is dense in $W^{1,p}(\Omega)$ for $1 \leq p < \infty$, i.e. for every

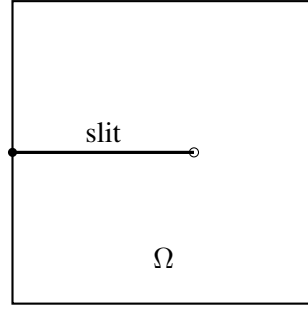


Figure 2.1: A slit domain: a square with a line segment removed from its interior.

$u \in W^{1,p}(\Omega)$ there exists a sequence $\{u_k\} \subset C^\infty(\overline{\Omega})$ such that

$$\|u_k - u\|_{W^{1,p}(\Omega)} \rightarrow 0 \quad \text{as } k \rightarrow \infty.$$



2.2 Integration by parts

2.2.1 One-dimensional case

By the fundamental theorem of calculus, if $f \in C^1([a, b])$, then

$$f(b) - f(a) = \int_a^b f'(s) ds = f|_a^b.$$

Apply this to $f = uv$, where $u, v \in C^1([a, b])$. Then

$$uv|_a^b = \int_a^b (uv)'(s) ds = \int_a^b u'(s)v(s) ds + \int_a^b u(s)v'(s) ds.$$

Hence the one-dimensional integration-by-parts formula

$$\int_a^b u'(s)v(s) ds = uv|_a^b - \int_a^b v'(s)u(s) ds.$$

2.2.2 Gauss theorem in higher dimensions

Let $\Omega \subset \mathbb{R}^d$ be a bounded domain and $f : \Omega \rightarrow \mathbb{R}^d$ a vector field. Gauss (divergence) theorem states that

$$\int_{\partial\Omega} f \cdot n dA = \int_{\Omega} \nabla \cdot f dx,$$

where $f \in C^1(\overline{\Omega})^d$ and

$$\nabla \cdot f = \sum_{i=1}^d \frac{\partial f_i}{\partial x_i}.$$

The quantity

$$\int_{\partial\Omega} 1 dA$$

is the $(d-1)$ -dimensional measure (surface area) of $\partial\Omega$.

Assume that $\partial\Omega$ is of class C^1 , namely that $\partial\Omega$ is the union of a finite number of patches U_i , $i = 1, \dots, k$, each of which can be mapped via an invertible C^1 function to a subset of $\mathbb{R}^{d-1}$.

2.2.3 Integration by parts in higher dimensions

Let $u : \Omega \rightarrow \mathbb{R}^d$ be a vector field with $u \in C^1(\overline{\Omega})^d$, and let $v : \Omega \rightarrow \mathbb{R}$ be a scalar function with $v \in C^1(\overline{\Omega})$. Apply Gauss theorem to the vector field

$$uv := (u_1v, \dots, u_dv) : \Omega \rightarrow \mathbb{R}^d.$$

We have

$$\int_{\partial\Omega} v u \cdot n \, dA = \int_{\Omega} \nabla \cdot (uv) \, dx.$$

Using the product rule,

$$\nabla \cdot (uv) = (\nabla \cdot u) v + u \cdot \nabla v,$$

we obtain the integration-by-parts formula

$$\int_{\Omega} u \cdot \nabla v \, dx = \int_{\partial\Omega} v(u \cdot n) \, dA - \int_{\Omega} (\nabla \cdot u) v \, dx. \quad (2.1)$$

Now choose a special vector field whose only nonzero component is the j th one:

$$F(x) = (0, \dots, 0, u(x), 0, \dots, 0),$$

so that $(\nabla \cdot F)(x) = \frac{\partial u}{\partial x_j}(x)$ and $F \cdot \nabla v = u \frac{\partial v}{\partial x_j}$. Applying (2.1) with F in place of u gives

$$\int_{\Omega} u \frac{\partial v}{\partial x_j} \, dx = \int_{\partial\Omega} v u n_j \, dA - \int_{\Omega} \frac{\partial u}{\partial x_j} v \, dx. \quad (2.2)$$

2.2.4 Smooth functions and weak derivatives

Let α be a multi-index with $|\alpha| \leq k$.

Theorem 2.6

If $u \in C^k(\overline{\Omega})$, then the weak derivative $D^\alpha u$ exists in $L^1_{loc}(\Omega)$ and coincides with the classical (strong) derivative.

Proof Let $\varphi \in C_c^\infty(\Omega)$ be a test function and assume $|\alpha| \geq 1$ (the case $|\alpha| = 0$ is trivial). Write $\alpha = \beta + e_j$ with $|\beta| = |\alpha| - 1$ and e_j the j th coordinate unit vector. Then

$$(-1)^{|\alpha|} \int_{\Omega} u D^\alpha \varphi \, dx = (-1)^{|\alpha|} \int_{\Omega} u \frac{\partial}{\partial x_j} D^\beta \varphi \, dx.$$

Apply the one-coordinate integration-by-parts formula (2.2) with $v = D^\beta \varphi$. Since φ has compact support in Ω , the boundary term vanishes and we get

$$(-1)^{|\alpha|} \int_{\Omega} u D^\alpha \varphi \, dx = (-1)^{|\alpha|-1} \int_{\Omega} \frac{\partial u}{\partial x_j} D^\beta \varphi \, dx.$$

Repeating this procedure $|\alpha|$ times (moving one derivative at a time from φ to u), we obtain

$$(-1)^{|\alpha|} \int_{\Omega} u D^\alpha \varphi \, dx = \int_{\Omega} D^\alpha u \varphi \, dx.$$

Thus the classical derivative $D^\alpha u$ is also the weak derivative. Since $u \in C^k(\overline{\Omega})$, we have $D^\alpha u \in C(\overline{\Omega})$ and in particular $D^\alpha u \in L^\infty(\Omega)$.

2.3 Introduction: Trace on a Square

Assume Ω has a continuous boundary so that $C^\infty(\overline{\Omega})$ is dense in $H^1(\Omega)$.

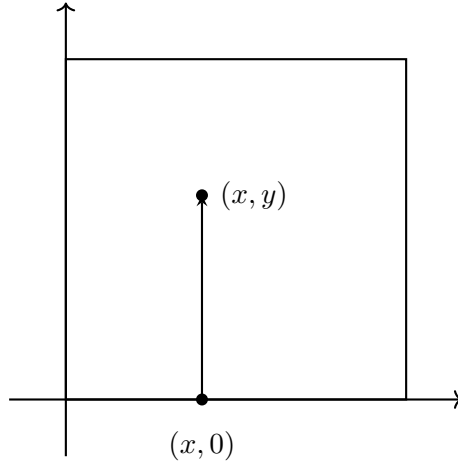
2.3.1 Special Case: $\Omega = (0, 1)^2$ 

Figure 2.2: Special case of the trace theorem

We start with 1D integration by parts (Fundamental Theorem of Calculus). Let $u \in C^1(\overline{\Omega})$. Consider the boundary at $y = 0$. By the Fundamental Theorem of Calculus:

$$u(x, 0) = - \int_0^y u_s(x, s) ds + u(x, y).$$

Recall the inequality $(a + b)^2 \leq 2(a^2 + b^2)$. Applying this:

$$|u(x, 0)|^2 \leq 2 \left(\left| \int_0^y u_s(x, s) ds \right|^2 + |u(x, y)|^2 \right).$$

Using Cauchy-Schwarz inequality on the integral term:

$$\begin{aligned} \left| \int_0^y u_s(x, s) ds \right|^2 &\leq \left(\int_0^y 1^2 ds \right) \left(\int_0^y |u_s(x, s)|^2 ds \right) \\ &= y \int_0^y |u_s(x, s)|^2 ds \\ &\leq 1 \cdot \int_0^1 |u_s(x, s)|^2 ds \quad (\text{since } y \leq 1). \end{aligned}$$

Substituting this back, we get:

$$|u(x, 0)|^2 \leq 2 \left(\int_0^1 |u_s(x, s)|^2 ds + |u(x, y)|^2 \right).$$

Now, integrate with respect to y over $[0, 1]$:

$$\begin{aligned} \int_0^1 |u(x, 0)|^2 dy &\leq \int_0^1 2 \left(\|u_s(x, \cdot)\|_{L^2(0,1)}^2 + |u(x, y)|^2 \right) dy \\ |u(x, 0)|^2 \cdot 1 &\leq 2 \|u_y(x, \cdot)\|_{L^2(0,1)}^2 + 2 \int_0^1 |u(x, y)|^2 dy. \end{aligned}$$

Next, integrate with respect to x over $[0, 1]$:

$$\begin{aligned} \int_0^1 |u(x, 0)|^2 dx &\leq 2 \int_0^1 \int_0^1 |u_y(x, y)|^2 dy dx + 2 \int_0^1 \int_0^1 |u(x, y)|^2 dy dx \\ \|u(\cdot, 0)\|_{L^2(0,1)}^2 &\leq 2 \left(\|u_y\|_{L^2(\Omega)}^2 + \|u\|_{L^2(\Omega)}^2 \right) \\ &\leq 2 \|u\|_{H^1(\Omega)}^2. \end{aligned}$$

Repeating this argument for each boundary segment (4 sides), we get:

$$\|u\|_{L^2(\partial\Omega)}^2 \leq 8\|u\|_{H^1(\Omega)}^2.$$

2.3.2 Density Argument

Since $C^\infty(\overline{\Omega})$ is dense in $H^1(\Omega)$, given $w \in H^1(\Omega)$, there exists a sequence $\{u_n\} \subset C^\infty(\overline{\Omega})$ such that $u_n \rightarrow w$ in $H^1(\Omega)$. Then $\{u_n|_{\partial\Omega}\}$ is Cauchy in $L^2(\partial\Omega)$:

$$\|u_n - u_m\|_{L^2(\partial\Omega)} \leq \sqrt{8}\|u_n - u_m\|_{H^1(\Omega)} \rightarrow 0. \quad (2.3)$$

Thus, $u_n|_{\partial\Omega}$ has a limit in $L^2(\partial\Omega)$, which we call the **trace** of w on $\partial\Omega$, denoted by $\gamma w = w|_{\partial\Omega}$.

2.4 General Case: Continuous Boundary

Theorem 2.7

Assume that Ω has a continuous boundary. The map $u \mapsto \gamma(u) = u|_{\partial\Omega}$ extends continuously from $W^{1,p}(\Omega)$ to $L^p(\partial\Omega)$ for $p \in [1, \infty)$.

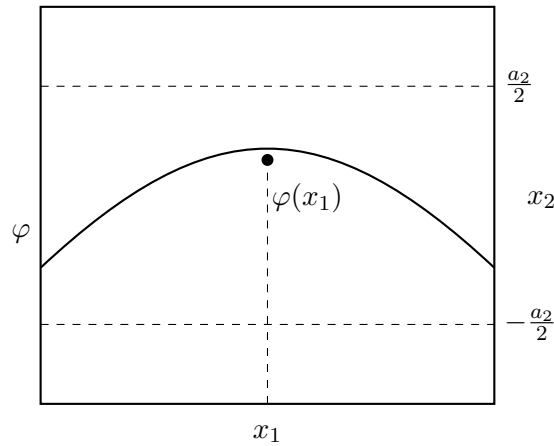


Figure 2.3: Generalized Boundary Patch

Proof Setup: Since $\partial\Omega$ is compact, we may reduce to a finite set of patches $V_1, \dots, V_N$ covering $\partial\Omega$. We use a local coordinate system. Let the boundary be locally described by a graph $x_2 = \varphi(x_1)$. Let the domain within the patch be defined by $-a_1 \leq x_1 \leq a_1$ and $-a_2 \leq x_2 \leq \varphi(x_1)$.

For $u \in C^1(\overline{\Omega})$, using the Fundamental Theorem of Calculus along the x_2 direction:

$$u(x_1, \varphi(x_1)) = \int_{x_2}^{\varphi(x_1)} u_s(x_1, s) ds + u(x_1, x_2). \quad (2.4)$$

Taking the absolute value:

$$|u(x_1, \varphi(x_1))| \leq \int_{x_2}^{\varphi(x_1)} |u_s(x_1, s)| ds + |u(x_1, x_2)|.$$

Using the inequality $(a + b)^p \leq 2^p(a^p + b^p)$ (for $p \geq 1$):

$$|u(x_1, \varphi(x_1))|^p \leq 2^p \left(\left| \int_{x_2}^{\varphi(x_1)} u_s(x_1, s) ds \right|^p + |u(x_1, x_2)|^p \right).$$

Apply Hölder's inequality to the integral term. Let q be the conjugate exponent of p ($1/p + 1/q = 1$).

$$\left| \int_{x_2}^{\varphi(x_1)} 1 \cdot u_s \, ds \right| \leq \|1\|_{L^q(x_2, \varphi_1(x_1))} \|u_s(x_1, s)\|_{L^p(x_2, \varphi_1(x_1))}$$

And

$$\|1\|_{L^q(x_2, \varphi(x_1))} = \begin{cases} 1 & \text{if } q = \infty \\ (\varphi(x_1) - x_2)^{\frac{1}{q}} \leq \left(\frac{3}{2}a_2\right)^{\frac{1}{q}} \leq C. \end{cases}$$

Note that the length is bounded by the size of the patch, say C . Thus,

$$|u(x_1, \varphi(x_1))|^p \leq C \left(\int_{-a_2}^{\varphi(x_1)} |u_s(x_1, s)|^p \, ds + |u(x_1, x_2)|^p \right).$$

Integration: Now integrate with respect to x_2 over the vertical segment $[-a_2, \varphi(x_1)]$. Note that the Left Hand Side (LHS), $|u(x_1, \varphi(x_1))|^p$, does not depend on x_2 .

$$\begin{aligned} \text{Integrated LHS} &= \int_{-a_2}^{\varphi(x_1)} |u(x_1, \varphi(x_1))|^p \, dx_2 \\ &= (\varphi(x_1) - (-a_2)) |u(x_1, \varphi(x_1))|^p \\ &\geq \frac{a_2}{2} |u(x_1, \varphi(x_1))|^p \quad (\text{assuming patch geometry is bounded away from 0}). \end{aligned}$$

Now integrate with respect to x_1 over $[-a_1, a_1]$:

$$\int_{-a_1}^{a_1} \frac{a_2}{2} |u(x_1, \varphi(x_1))|^p \, dx_1 \leq C \int_{-a_1}^{a_1} \int_{-a_2}^{\varphi(x_1)} (|u_s|^p + |u|^p) \, dx_2 \, dx_1.$$

This simplifies to:

$$C_0 \|u\|_{L^p(\partial\Omega \cap V)}^p \leq C \left(\|u_{x_2}\|_{L^p(\Omega \cap V)}^p + \|u\|_{L^p(\Omega \cap V)}^p \right).$$

In standard coordinates, this implies:

$$\|u\|_{L^p(\partial\Omega \cap V)}^p \leq C \|u\|_{W^{1,p}(\Omega \cap V)}^p.$$

Conclusion: Summing up for all finite number of patches covering the boundary, there holds:

$$\|u\|_{L^p(\partial\Omega)} \leq C \|u\|_{W^{1,p}(\Omega)}.$$

Chapter 3 Finite Element Space

3.1 The Finite Element

3.1.1 Admissible Triangulation ($\mathbb{R}^2$)

Assume that Ω is polygonal in 2D or polyhedral in 3D, i.e. $\Omega = \left(\bigcup_{\tau_i \in \mathcal{J}} \tau_i \right)^\circ$

1. τ_i is a (closed) rectangle or triangle. ($\Omega \subset \mathbb{R}^2$)
2. τ_i is a (closed) brick shaped element on. a tetrahedra ($\Omega \subseteq \mathbb{R}^3$)
3. If τ_i and τ_j are distinct elements in $\mathcal{J}$, then.
 - (a). $\tau_i \cap \tau_j = \emptyset$;
 - (b). or $\tau_i \cap \tau_j$ is a vertex or an edge of both τ_i and τ_j . ($\Omega \subseteq \mathbb{R}^2$)
 - (c). $\tau_i \cap \tau_j$ is a vertex, edge, face of both τ_i and τ_j . ($\Omega \subseteq \mathbb{R}^3$).

Remark Hanging node is not allowed.

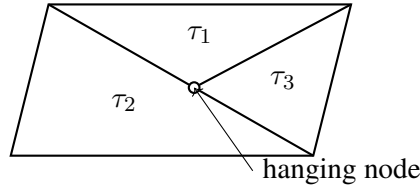


Figure 3.1: Hanging node

Theorem 3.1

Suppose that $\mathcal{J}$ is a triangulation and Ω satisfying $\mu(\tau_i \cap \tau_j) = 0$ if τ_i and τ_j are distinct. Suppose that $f|_{\tau_i}$ is smooth for each $\tau_i \in \mathcal{J}$. Then $f \in H^k(\Omega)$ iff $f^{(i)}$ is continuous for $j = 0, 1, \dots, k-1$.

Remark DoF: Degrees of freedom allow for local basis, global basis, interpolation and continuity across element interface.

3.1.2 General finite element

General finite element (K, P, Σ) .

- K - region
- P - shape function $\subseteq C^\ell(K)$ with finite dimension m
- $\Sigma = \{\ell_1, \dots, \ell_m\}$ linear functionals which are unisolvent on P

Definition 3.1 (Unisolvent)

$\{\ell_1, \dots, \ell_m\}$ linear functionals on P (dimension m) is unisolvent iff the only solution q to $\ell_i(q) = 0 (i = 1, \dots, m)$ is $q = 0$.

Example 3.1 K = triangle in $\mathbb{R}^2$

$$P = \mathbb{P}^k = \left\{ \sum_{i=0}^k \sum_{j=0}^{k-i} c_{ij} x_1^i x_2^j, c_{ij} \in \mathbb{R} \right\}, \quad \dim P = 1 + 2 + \dots + k + 1 = \frac{(k+1)(k+2)}{2}$$

$$\Sigma = \{ \ell_{ij}(q) = q(x_{ij}), i = 0, \dots, k, j = 0, \dots, k-i, x_{ij} \in K \}.$$

We need to define $\{x_{ij}\}$.

- Case 1: $k = 0$, $\mathbb{P}^0 = \text{constant functions}$, $\dim(\mathbb{P}^0) = 1$.

$$x_{0,0} = \frac{V_1 + V_2 + V_3}{3} \text{ (barycenter of } \tau), \quad \ell_{00}(q) = q(x_{00}).$$

We cannot use this element to get a nontrivial subspace of $H^1(\Omega)$. But it works for approximation in $L^2(\Omega)$.

- Case 2: $k = 1$, $\mathbb{P}^1 = \{ \alpha + \beta x_1 + \gamma x_2, \alpha, \beta, \gamma \in \mathbb{R} \}$, $\dim(\mathbb{P}^1) = 3$.

$$\Sigma = \{ \ell_i(q) = q(v_i), i = 1, 2, 3 \}.$$

Question: Is this a unisolvent set of functionals for $\mathbb{P}^1$?

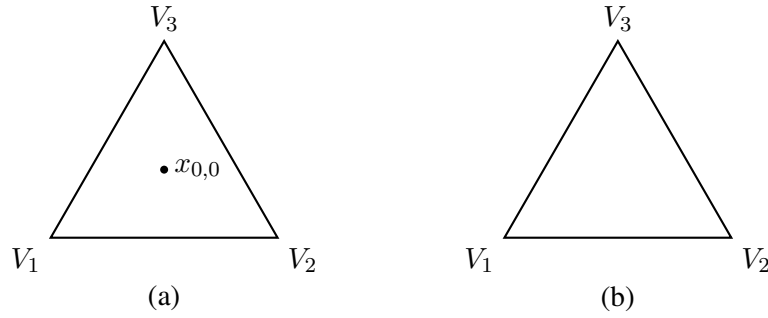


Figure 3.2: finite element for example(3.1)

3.1.3 Reference Element

Lemma 3.1

If $f \in \mathbb{P}^k$, and $\ell(S) = v_1 + (v_2 - v_1)s$, then $\ell(0) = v_1, \ell(1) = v_2$. Define $q_k(s) = f(\ell(s))$. Then q_k is a polynomial of degree k .



Proof $\ell(s) = \begin{pmatrix} v_{11} + (v_{21} - v_{11})s \\ v_{12} + (v_{22} - v_{12})s \end{pmatrix}$

$$f(\ell(s)) = \sum_{0 \leq i+j \leq k} c_{ij} (v_{1,1} + (v_{2,1} - v_{1,1})s)^i (v_{1,2} + (v_{2,2} - v_{1,2})s)^j$$

This is the polynomial in s of degree k .

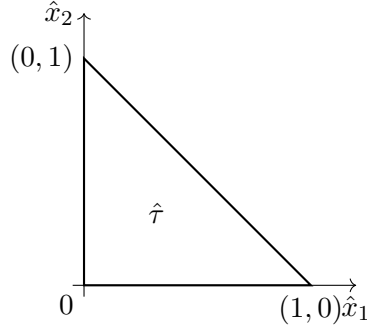


Figure 3.3: Reference element

Now if $p \in \mathbb{P}^1$ satisfies $p(v_i) = 0$ where $i = 0, 1, 2$, $v_0 = (0, 0)$, $v_1 = (1, 0)$, $v_2 = (0, 1)$.

Restricted to the edge $\hat{x}_2 = 0$ is linear in x_1 , which vanishes at 0 and 1. Therefore, it vanishes on the edge $x_2 = 0$. So $p = \alpha + \beta x_1 + \gamma x_2 = 0$ when $x_2 = 0$. Thus $\alpha = \beta = 0$. Finally, $p(v_2) = 0 = \gamma x_2$, we get $\gamma = 0$ i.e. $p = 0$. we get the unisolvence.

Definition 3.2

An affine mapping of $\mathbb{R}^d$ is one of the form

$$Bx = C_0 + Mx, \quad x \in \mathbb{R}^d.$$

Here $C_0 \in \mathbb{R}^d$, $M \in \mathbb{R}^{d \times d}$.



Remark The transform is invertible if M is nonsingular.

$$\begin{aligned} B^{-1}y &= M^{-1}(y - C_0) \\ &= -M^{-1}C_0 + M^{-1}y \end{aligned}$$

and B^{-1} is also affine.

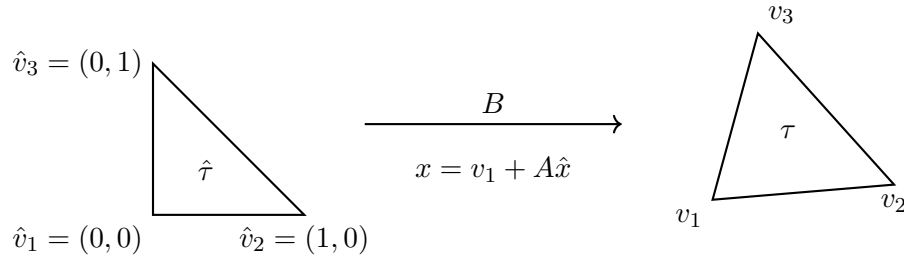


Figure 3.4: Affine mapping

In our case, $x = B\hat{x} = v_1 + \begin{pmatrix} v_2 - v_1 & v_3 - v_1 \end{pmatrix} \hat{x}$ and there holds $B(\hat{v}_i) = v_i$.

Exercise 3.1 B is invertible if v_1, v_2, v_3 are not on a line. (or τ has positive area).

Proposition 3.1

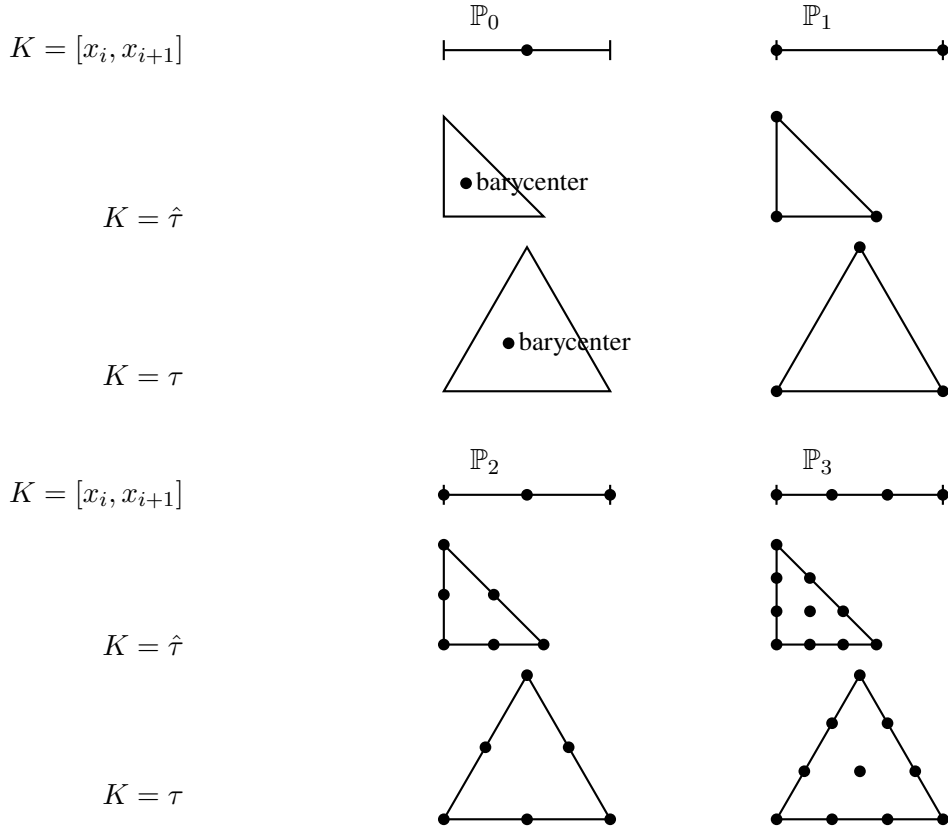
B maps $\hat{\tau}$ 1:1 onto τ .

**Lemma 3.2**

B 1 : 1 map from $\mathbb{P}^k(\hat{\tau})$ to $\mathbb{P}^k(\tau)$. (map 0 to 0 one-to-one)



3.2 Lagrange Element



$$\mathbb{P}_4, \mathbb{P}_5, \dots, \mathbb{P}^k = \left\{ \sum_{0 \leq i+j \leq k} C_{ij} x_1^i x_2^j \right\},$$

$$\dim \mathbb{P}^k = (k+1) + k + \dots + 1 = \frac{1}{2}(k+1)(k+2).$$

Proposition 3.2

$\mathbb{P}^k$ Lagrange elements are unisolvent (2D Triangle case).

Proof $B : \hat{\tau} \mapsto \tau$, is an affine and invertible when τ is nondegenerate. B maps the $(\mathbb{P}^k)$ nodes $\hat{N}_k$ of $\hat{\tau}$ onto N_k of τ . If the nodes of $\hat{N}_k$ are unisolvent, then the nodes of N_k are unisolvent, namely if $p \in \mathbb{P}^k$ and satisfies $P(x_j) = 0, x_j \in N_k$, then $p = 0$. Consider $q(\hat{x}) = p(B(\hat{x}))$, since $p \mapsto p \circ B$ is a bijection of $\mathbb{P}^k$ onto $\mathbb{P}^k$ and since B maps nodes to nodes and p vanishes on the nodes of N_k , q vanishes on the nodes $\hat{N}_k$. The unisolvence of $\hat{N}_k$ implies that $q = 0$, hence $p = 0$. Therefore, it suffices to verify the result only on the reference triangle. Next we show by induction on k .

$k = 0, \mathbb{P}^k = \text{constants}$ DONE!

Now we assume unisolvence holds for $\mathbb{P}^{k-1}$ in $\hat{\tau}$, we need only show the case on $\mathbb{P}^k$ for $\hat{N}_k$.

Assume $p(x_i) = 0, x_i \in \hat{N}_k$, consider $q = p|_{\hat{x}_2=0} = q(\hat{x}_1)$. So $q(\hat{x}_1) \in \mathbb{P}^k(0, 1)$, and q vanishes at $k+1$ nodes. Thus $q = p|_{\hat{x}_2=0} = 0$ (1D case). Let $p = \sum_{0 \leq i+j \leq k} c_{ij} \hat{x}_1^i \hat{x}_2^j$. Thus

$$p|_{\hat{x}_2=0} = \sum_{i=0}^k c_{i0} \hat{x}_1^i = 0 \Rightarrow c_{i0} = 0, \quad i = 0 \dots, k.$$

$$\begin{aligned}
p &= \sum_{i=0}^k \sum_{j=1}^{k-i} c_{ij} \hat{x}_1^i \hat{x}_2^j = \hat{x}_2 \sum_{i=0}^k \sum_{j=1}^{k-i} c_{ij} \hat{x}_1^i \hat{x}_2^{j-1} \\
&= \hat{x}_2 \sum_{i=0}^k \sum_{j=0}^{k-i-1} c_{i,j+1} \hat{x}_1^i \hat{x}_2^j = \hat{x}_2 \sum_{0 \leq i+j \leq k-1} c_{i,j+1} \hat{x}_1^i \hat{x}_2^j \\
&= \hat{x}_2 \tilde{q}(\hat{x}_1, \hat{x}_2).
\end{aligned}$$

and $\tilde{q}(\hat{x}_1, \hat{x}_2) \in \mathbb{P}^{k-1}$. As $\hat{x}_2 \neq 0$ on $N_{k-1}(\hat{\tau})$, $\tilde{q}(x_j) = 0$, $x_j \in N_{k-1}(\hat{\tau})$. By the induction assumption, $\tilde{q}(\hat{x}) = 0$, we have $p(\hat{x}) = 0$, DONE!

Remark The proof easily extends to the case of $\mathbb{R}^d$. (an additional induction over)

Let $S_h = \{\varphi \in C(\bar{\Omega}), \varphi|_{\tau} \in \mathbb{P}^k, \tau \in \mathcal{T}\} \subset H^1(\Omega)$, where $\mathcal{T}$ is an admissible triangulation of Ω .

Other characterization

$\mathcal{N}_k(\mathcal{T}) = \bigcup_{\tau \in \mathcal{T}} \mathcal{N}(\tau)$, where $\mathcal{N}(\tau)$ denotes the set of nodes(DoFs) in τ . We set $\widetilde{S}_h$ to be the collection of all piecewise $\mathbb{P}^k$ functions determined by these nodes. So given $v(x_j)$, $x_j \in \mathcal{N}$. There exists a unique $u \in \widetilde{S}_h$ satisfying $u(x_j) = v(x_j)$, $x_j \in \mathcal{N}$.

Proposition 3.3

$$\widetilde{S}_h = S_h.$$

Proof

- $\widetilde{S}_h \subset S_h$, we need to show that $\varphi \in \widetilde{S}_h$ is continuous. Let $\varphi_1 = \varphi|_{\tau_1}$ and $\varphi_2 = \varphi|_{\tau_2}$. So $\varphi_2|_e$ and $\varphi_1|_e$ are 1D polynomials of degree k , which share the same values at the $k+1$ distinct nodes on e . Both must equal the unique polynomial interpolating the $k+1$ values on the edge and hence $\varphi_2|_e = \varphi_1|_e$, namely φ is continuous on e .
- $S_h \subset \widetilde{S}_h$ is clear since given $\varphi \in S_h$, we set $\tilde{\varphi}$ by $\tilde{\varphi}(x_i) = \varphi(x_i)$ ($x_i \in \mathcal{N}$) as both φ and $\tilde{\varphi}$ are in $\mathbb{P}^k$ on each triangle and agree on the nodes of the triangle. Thus $\tilde{\varphi} = \varphi$ ($\forall \tau \in \mathcal{T}$), we get $\varphi \in \widetilde{S}_h$.

We then have the finite element basis $\varphi_i \in S_h$ such that.

$$\varphi_i(x_j) = \delta_{ij}, x_j \in \mathcal{N}$$

and define the interpolation operator

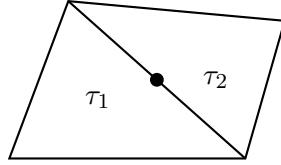
$$\mathcal{I}_h(u) = \sum_{x_i \in \mathcal{N}} u(x_i) \varphi_i(x).$$

Example 3.2 $K = \tau$ (triangle), $P = \mathbb{P}^1$ (linear). $\Sigma = \{\ell_i(p) = p(m_i), i = 1, 2, 3\}$ Is Σ unisolvent?



Figure 3.5: $\mathbb{P}^1$ element on a triangle

As in the earlier proof, this will be unisolvent if is unisolvent on $\hat{\tau}$.



Exercise 3.2 check that when plug in the points in $p = \alpha + \beta \hat{x}_1 + \gamma \hat{x}_2$, the corresponding 3×3 matrix is nonsingular.

N = midpoints of edges of $\tau, \tau \in \mathcal{T}$. S_h = Set of piecewise linear functions determined by the nodes.

$\varphi \in S_h, \varphi_1 = \varphi|_{\tau_1}, \varphi_2 = \varphi|_{\tau_2}$. This is not enough to impose continuity. The resulting space is not a subset of $H^1(\Omega)$. But it is useful to solve other problems. We call such finite element as “Crouvix - Raviart elements”.

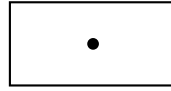
Example 3.3

K = rectangle.

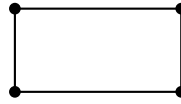
$$P = \mathbb{Q}^k = \left\{ \sum_{i,j=0}^k c_{ij} x_1^i x_2^j \right\} \subset \mathbb{P}^{2k} \quad \dim = (k+1)^2.$$

$$\mathbb{P}^k \subseteq \mathbb{Q}^k$$

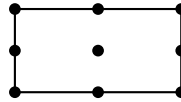
$k = 0$



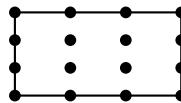
$k = 1$: bilinears



$k = 2$: bi-quadratics



$k = 3$



etc.

Figure 3.6: Tensor product element

Consider B an affine mapping, $\hat{\tau} = (0, 1)^2$. $B : \hat{\tau} \rightarrow \tau$.

$$p \in \mathbb{Q}^k \rightarrow q = p \circ B \in \mathbb{Q}^{2k}$$

$$p = \sum_{i,j=0}^k c_{ij} x_1^i x_2^j \Rightarrow p \circ B = \sum_{i,j=0}^k c_{ij} (b_{11}\hat{x}_1 + b_{12}\hat{x}_2)^i (b_{21}\hat{x}_1 + b_{22}\hat{x}_2)^j$$

So B does not necessarily map $\mathbb{Q}^k \rightarrow \mathbb{Q}^k$.

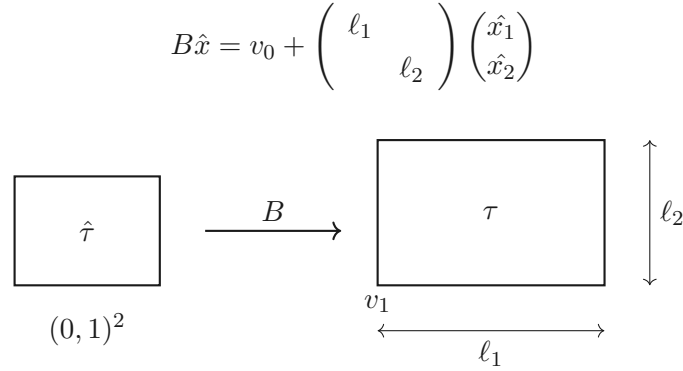
Proposition 3.4

The nodes are unisolvent.



Notation: τ is a rectangle and $\mathcal{N}_k(\tau)$ is the set of nodes for $\mathbb{Q}^k$ on τ .

Proof Given τ , we can map $\hat{\tau}$ onto τ by translating and dialation in each direction.



This transformation $P \rightarrow P \circ B$ maps $\mathbb{Q}^k$ onto $\mathbb{Q}^k$. This means that it suffices to show the unisolvence for the reference $\hat{\tau} = (0, 1)^2$ and implies unisolvence for all τ .

Do induction on k . For $k = 0$, DONE!

Assume it holds for $k - 1$. (all τ). We need only show that it works for the reference $\hat{\tau}$. If $p \in \mathbb{Q}^k$ then $p|_{\text{edge}}$ of τ is a polynomial $q \in \mathbb{P}^k(1D)$.

Assume $p \in \mathbb{Q}^k$ and $p(x_j) = 0$ for $x_j \in \mathcal{N}_k(\hat{\tau})$, $p|_{x_2=0} \in \mathbb{P}^k(1D)$ and vanishes at $k + 1$ distinct nodes $\Rightarrow p = 0$ on $x_2 = 0$. Also $p = 0$ on $x_1 = 0$. Then, it follows that

$$p = x_1 x_2 \sum_{i,j=1}^k C_{ij} x_1^{i-1} x_2^{j-1} \in \mathbb{Q}^{k-1}.$$

i.e $p = x_1 x_2 q$, with $q \in \mathbb{Q}^{k-1}$. And q vanishes for $x_j \in \mathcal{N}_{k-1}(\hat{\tau})$, By induction $q = 0$, so $p = 0$.

3.3 $H^2(\Omega)$ conforming elements

In this section, we discuss about finite element spaces that are subspace of $H^2(\Omega)$. As the previous lemma. for H^2 -conforming elements, we require that the finite element functions are not only continuous, but its gradient are also continuous in the entire region.

$$K = (x_0, x_1), \quad P = \mathbb{P}^3, \quad \Sigma = \{\ell_0(p) = p(x_0), \ell_1(p) = p(x_1), \ell_2(p) = p'(x_0), \ell_3(p) = p'(x_1)\}$$

Proposition 3.5 (unisolvence)

There is a unique cubic satisfying the interpolation problem: $p \in \mathbb{P}^3$ so that

$$p(x_i) = f_i, \quad i = 0, 1 \quad \text{and} \quad p'(x_i) = g_i, \quad i = 0, 1.$$

Note that x_0 , and x_1 are distinct.

Proof We need only to show that the only solution to $p(x_i) = p'(x_i) = 0$ in $\mathbb{P}^3$ is the zero polynomial. if $p(x_i) = p'(x_i) = 0$, then $(x - x_i)^2$ divides p . So $(x - x_0)^2 (x - x_1)^2$ divides p , but p is cubic. So p must be 0.

Now we consider the two-dimensional case. The corresponding finite element space is called Argyris

elements:

$K = \text{triangle.}$

$$P = \mathbb{P}^5 = \left\{ \sum_{0 \leq i+j \leq 5} C_{ij} x_1^i x_2^j \right\}$$

$$\Sigma = \left\{ D^\alpha P(v_i), i = 1, 2, 3, |\alpha| \leq 2, \frac{\partial P}{\partial \mathbf{n}}(m_i), i = 1, 2, 3 \right\}$$

$$\dim(\Sigma) = 3 \times 6 + 3 = 21 = \dim(\mathbb{P}^5).$$

Proposition 3.6

The nodes are unsolvent.



Remark The normal derivative does not map to the that in the reference triangle by affine mapping.

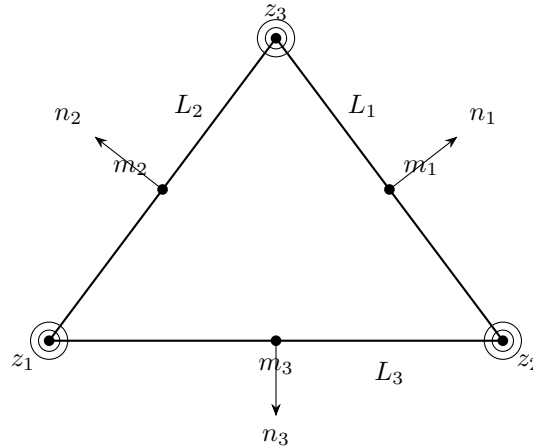
Lemma 3.3

1. In the one dimensional case, consider $K = [x_0, x_1]$, $p = \mathbb{P}^4$ and $\Sigma = \{P(x_i), P'(x_i), i = 0, 1 \text{ and } P(\frac{x_0+x_1}{2})\}$, we can show that Σ is unsolvent for $\mathbb{P}^4$. (Exercise).
2. If $\Sigma = \{P(x_i), P'(x_i), P''(x_i) \text{ with } i = 0, 1\}$. Then Σ is unsolvent for $\mathbb{P}^5$. (Exercise).



Lemma 3.4

Let $B : \hat{\tau} \rightarrow \tau$ be an affine map with $B(\hat{e}_i) = e_i$. If $D^\alpha f|_{e_i} = 0, |\alpha| \leq 1$, then $D^\alpha(f \circ B)|_{\hat{e}_i} = 0, |\alpha| \leq 1$. (Exercise; chain rule).



Proof [Proof of the proposition]

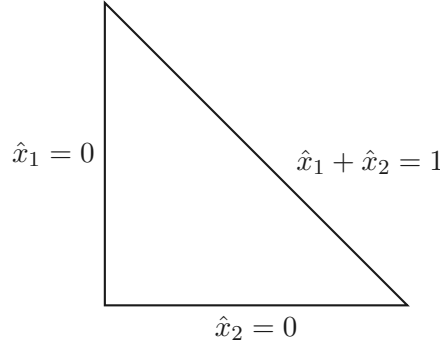
Assume $P \in \mathbb{P}^5$ and vanishes at the nodes of τ . Then: $P(v_i) = 0$, $\frac{\partial P}{\partial \tau}(v_i) = 0$, and $\frac{\partial^2 P}{\partial \tau^2}(v_i) = 0$ for $i = 1, 2$. As $P|_e$ is a polynomial of degree 5. Using Lemma 3.3 implies that $P = 0$ on e . and $P_\tau = \nabla P \cdot \tau$ vanishes on e .

On the other hand, $\frac{\partial P}{\partial \mathbf{n}}$ is in $\mathbb{P}^{k-1}$ (Exercise). $\frac{\partial P}{\partial \mathbf{n}}|_e$ is in $\mathbb{P}^{k-1}$. and $\frac{\partial P}{\partial \mathbf{n}}(m) = 0$, $\frac{\partial P}{\partial \mathbf{n}}(v_i) = 0, i = 1, 2$. and $\frac{\partial}{\partial \tau} \left(\frac{\partial p}{\partial \mathbf{n}} \right)(v_i) = 0, i = 1, 2$ (combination of free 2nd derivative). By Lemme 3.3, $\frac{\partial P}{\partial \mathbf{n}} = 0$ on e . So p and its first derivative vanishes on each edge. Lemma 3.4 implies that $q = P \circ B$ and its derivatives vanishes on each

edge of $\hat{\tau}$. $q = \sum_{i+j=k} a_j \hat{x}_1^i \hat{x}_2^j$. So q vanishes on $\hat{x}_2 = 0$.

$$\Rightarrow C_{i,0} = 0, \quad i = 0, \dots, k.$$

$$\frac{\partial P}{\partial \hat{x}_2} = 0 \Rightarrow C_{i1} = 0, \quad i = 0, \dots, k-1.$$



Thus $\hat{x}_2^2$ divides q . Similarly $\hat{x}_1^2$ divides q . Therefore $q = \hat{x}_1^2 \hat{x}_2^2 (\alpha + \beta \hat{x}_1 + \gamma \hat{x}_2)$. Apply to $\hat{x}_1 + \hat{x}_2 = 1 \Rightarrow (1 - \hat{x}_1 - \hat{x}_2)^2$ divides $q \Rightarrow \alpha = \beta = \gamma = 0$.

Theorem 3.2

Let $\mathcal{T}$ be a triangulation of $\Omega \subseteq \mathbb{R}^2$. The Argyris finite element space is in $C^1(\bar{\Omega})$. i.e. it is a subspace of $H^2(\Omega)$. ($H^2(\Omega)$ -conforming)



Proof If triangles meet at a vertex, then the function and its 4th derivative values are the same at the vertex.

Suppose we have a piecewise quintic on τ_1 and τ_2 which share the DoFs on V_1, V_2 , and m .

Let $p|_{\tau_1} = p_1, p|_{\tau_2} = p_2$. Since any first derivative is a linear combination of p_{x_1} and p_{x_2} .

So $\frac{\partial p_1}{\partial \tau}(v_i) = \frac{\partial p_2}{\partial \tau}(v_i), i = 1, 2$. Similarly. $\frac{\partial^2 p}{\partial x_1^2}, \frac{\partial^2 p}{\partial x_1 \partial x_2}$ and $\frac{\partial^2 p}{\partial x_2^2}$ imply that

$$\begin{aligned} \frac{\partial^2 p_1}{\partial \tau^2}(v_i) &= \frac{\partial^2 p_2}{\partial \tau^2}(v_i) \quad i = 1, 2 \\ \frac{\partial^2 p_1}{\partial \tau \partial \mathbf{n}}(v_i) &= \frac{\partial^2 p_2}{\partial \tau \partial \mathbf{n}}(v_i) \quad i = 1, 2. \end{aligned}$$

Now let $q_j = p_j|_e \in \mathbb{P}^5(1D)$. $j = 1, 2$. We have

$$\begin{aligned} q_1(v_i) &= q_2(v_i) \quad i = 1, 2 \\ q'_1(v_i) &= q'_2(v_i) \quad i = 1, 2. \\ q''_1(v_i) &= q''_2(v_i) \quad i = 1, 2. \end{aligned}$$

These DoFs are unisolvent on $\mathbb{P}^5(1D) \Rightarrow q_1 = q_2$.

So the function is continuous.

Next we have to check its derivatives are continuous. It suffices to check normal derivatives on e are continuous. Let $q_i = \frac{\partial p_i}{\partial \mathbf{n}}|_e \in \mathbb{P}^4(1D), i = 1, 2$. So

$$\begin{aligned} q_1(v_i) &= q_2(v_i) \quad i = 1, 2 \\ \frac{\partial^2 p_1}{\partial \tau \partial \mathbf{n}}(v_i) &= \frac{\partial q_1}{\partial \tau}(v_i) = \frac{\partial q_2}{\partial \tau}(v_i) = \frac{\partial^2 p_2}{\partial \tau \partial \mathbf{n}}(v_i). \\ q_1(m) &= q_2(m). \end{aligned}$$

By Lemma 3.3, we have $q_1 = q_2$. So the 1st derivative of f is continuous.

Chapter 4 Finite Element Method

4.1 Finite element approximation properties

Consider the reference triangulation element $\hat{\tau}$ in $\mathbb{R}^2$.

We shall show that

$$\inf_{p \in \mathbb{P}^0} \|u - p\|_{H^1(\hat{\tau})} \leq C |u|_{H^1(\hat{\tau})} = C \left(\int_{\hat{\tau}} |\nabla u|^2 dx \right)^{1/2}.$$

C doesn't depend on u .

$$\begin{aligned} \|u - p\|_{H^1(\hat{\tau})}^2 &= \|u - p\|_{L^2(\hat{\tau})}^2 + \|\nabla(u - p)\|_{L^2(\hat{\tau})}^2 \\ &= \|u - p\|_{L^2(\hat{\tau})}^2 + |u|_{H^1(\hat{\tau})}^2 \end{aligned}$$

We need only to bound

$$\inf_{p \in \mathbb{P}^0} \|u - p\|_{L^2(\hat{\tau})} \leq c |u|_{H^1(\hat{\tau})}.$$

Next we shall show that for $u \in C^\infty(\hat{\tau})$, there holds

$$\|u - \bar{u}\|_{L^2(\hat{\tau})} \leq c |u|_{H^1(\hat{\tau})}.$$

with

$$\bar{u} = \frac{1}{|\hat{\tau}|} \int_{\hat{\tau}} u dx = 2 \int_{\hat{\tau}} u dx = \oint_{\hat{\tau}} u dx \quad (\text{average}).$$

$$\begin{aligned} u(x) &= u(x) - u(x_1, y_2) + u(x_1, y_2) - u(y) + u(y) \\ &= \underbrace{\int_{y_2}^{x_2} u_s(x_1, s) ds + \int_{y_1}^{x_1} u_t(t, y_2) dt}_{J(x, y)} + u(y). \end{aligned}$$

Now we integrate y over $\hat{\tau}$:

$$\frac{1}{2} u(x) = \int_{\hat{\tau}} J(x, y) dy + \int_{\hat{\tau}} u(y) dy.$$

Thus

$$\begin{aligned} u(x) &= 2 \int_{\hat{\tau}} J(x, y) dy + \bar{u}. \\ \Rightarrow u(x) - \bar{u} &= 2 \int_{\hat{\tau}} J(x, y) dy. \end{aligned}$$

Square and integrate over x in $\hat{\tau}$ to get

$$\|u - \bar{u}\|_{L^2(\hat{\tau})}^2 = \int_{\hat{\tau}} |u(x) - \bar{u}|^2 dx = 4 \int_{\hat{\tau}} \left(\int_{\hat{\tau}} J(x, y) dy \right)^2 dx.$$

Bound the right-hand side using $(a + b)^2 \leq 2(a^2 + b^2)$ and Schwarz inequality. (Exercise).

$$\|u - \bar{u}\|_{L^2(\hat{\tau})}^2 \leq c |u|_{H^1(\hat{\tau})}^2$$

Lemma 4.1 (Deny-Lions)

Let B be connected bounded domain with a Lipschitz continuous boundary, then there exists a constant

C , such that.

$$\inf_{p \in \mathbb{P}^k} \|u - p\|_{H^{k+1}(B)} \leq C|u|_{H^{k+1}(B)} \equiv C \left(\sum_{|\alpha|=k+1} \|D^\alpha u\|_{L^2(B)}^2 \right)^{1/2}.$$



Remark This lemma sometimes is called the Bramble-Hilbert lemma.

Sketch of proof:

Consider $p \in \mathbb{P}^k$, $\ell_{ij}(p) = c_{ij}$ where $p = \sum_{0 \leq i+j \leq k} C_{ij} x_1^i x_2^j$. So ℓ_{ij} is a bonded linear functional on $\mathbb{P}^k$ (a finite dimensional space). By the Hahn-Banach thm, ℓ_{ij} can be extended to the linear functional on $H^{k+1}(B)$. We will show that

$$\|v\|_{H^{k+1}(B)} \leq C(B) \left\{ |v|_{H^{k+1}(B)} + \sum_{0 \leq i+j \leq k} |\ell_{ij}(v)| \right\} \quad (4.1)$$

If we verify equation 4.1, then we choose

$$q = \sum_{0 \leq i+j \leq k} \ell_{ij}(v) \cdot x_1^i x_2^j \in \mathbb{P}^k$$

So that

$$\inf_{p \in \mathbb{P}^k} \|v - p\|_{H^{k+1}(B)} \leq \|v - q\|_{H^{k+1}(B)}$$

Note that $\ell_{ij}(v - q) = \ell_{ij}(v) - \ell_{ij}(q) = 0$. By equation (4.1), we have

$$\|v - q\|_{H^{k+1}(B)} \leq C(B)|v - q|_{H^{k+1}(B)} = C(B)|v|_{H^{k+1}(B)}$$

(Since $D^\alpha q = 0$, for $|\alpha| = k + 1$.)

We already had shown that (4.1) implied the Deny-Lion Lemma. Suppose that (4.1) does not hold. This imply that there exists v_n with $\|v_n\|_{H^{k+1}(B)} = 1$ and

$$\left\{ |v_n|_{H^{k+1}(B)} + \sum_{0 \leq i+j \leq k} |\ell_{ij}(v_n)| \right\} < \frac{1}{n} \quad (4.2)$$

The compactness of $H^1(B)$ in $L^2(B)$ implies the compactness of $H^{k+1}(B)$ in $H^k(B)$. This means that there is a subsequence of $\{v_n\}$ which converges in $H^k(B)$. We still denote the subsequence by $\{v_n\}$ which still satisfies (4.2).

(4.2) implies the sequence is Cauchy in $H^{k+1}(B)$ and v_n convergent in $H^k(B)$ implies Cauchy in the $H^k(B)$ -norm $\Rightarrow$ convergent in $H^{k+1}(B)$, i.e, there exists $v \in H^{k+1}(B)$ with $v_n \rightarrow v$ in $H^{k+1}(B)$. Taking the limit in (4.1) implies $|v|_{H^{k+1}(B)} = 0 \Rightarrow D^\alpha v = 0$, $|\alpha| = k + 1$. So for functions in Sobolev spaces $D^\alpha v = 0$ for $|\alpha| = k + 1$ on a connected domain $B \Rightarrow v \in \mathbb{P}^k$. $v = \sum \ell_{ij}(v) x_1^i x_2^j$, $0 \leq i + j \leq k$. and taking the limit in (4.1) implies $\sum_{0 \leq i+j \leq k} |\ell_{ij}(v)| = 0 \Rightarrow v = 0$. This is a contradiction since $\|v_n\|_{H^{k+1}(B)} = 1$ cannot converge in H^{k+1} to a function with norm 0.

Remark This is also called the Bramble-Hilbert Lemma.

Definition 4.1

$\ell : V \rightarrow \mathbb{R}$ is sublinear and bounded if

1. $|\ell(v + w)| \leq |\ell(v)| + |\ell(w)|$, $\forall v, w \in V$
2. $|\ell(\alpha v)| = |\alpha| |\ell(v)|$.
3. $|\ell(v)| \leq c \|v\|_V$. (Bounded)



Lemma 4.2 (Bramble - Hilbert Lemma)

Assume B is bounded connected domain with Lipschitz domain. Let q be a bounded sublinear functional satisfying

$$q(p) = 0, \quad \forall p \in \mathbb{P}^k \text{ on } H^{k+1}(B).$$

Then there exists $C(B)$ such that

$$|q(v)| \leq C(B) \|q\| \|v\|_{H^{k+1}(B)}.$$

Here $\|q\| = \sup_{v \in H^{k+1}(B)} \frac{|q(v)|}{\|v\|_{H^{k+1}(B)}}.$



Example 4.1 $q(v) = \|v - \hat{\mathcal{I}}(v)\|_{H^{k+1}(B)}$ where $\hat{\mathcal{I}} : H^{k+1}(B) \rightarrow \mathbb{P}^k(B)$ and satisfies $\hat{\mathcal{I}}(p) = p$ for all $p \in \mathbb{P}^k$. $\hat{\mathcal{I}}$ is linear and bounded. And $q(p) = \|p - \hat{\mathcal{I}}p\|_{H^{k+1}} = 0 \quad \forall p \in \mathbb{P}^k$.

$$\begin{aligned} |q(v+w)| &= \|(v+w) - \hat{\mathcal{I}}(v+w)\|_{H^{k+1}(B)} \\ &\leq \|v - \hat{\mathcal{I}}(v)\|_{H^{k+1}(B)} + \|w - \hat{\mathcal{I}}(w)\|_{H^{k+1}(B)} \\ &= |q(v)| + |q(w)|. \end{aligned}$$

Proof [Bramble - Hilbert Lemma] $v \in H^{k+1}(B)$ and $p \in \mathbb{P}^k$

$$\begin{aligned} |q(v)| &= |q(v - p + p)| \\ &\leq |q(v - p)| + |q(p)| \\ &= |q(v - p)| \\ &\leq \|q\| \cdot \|v - p\|_{H^{k+1}(B)} \\ &\leq \|q\| \cdot c(B) \cdot |v|_{H^{k+1}(B)} \quad (\text{Deny-Lions}) \end{aligned}$$

4.2 Interpolation error on triangular

Let h_τ = diameter of τ = length of the largest edge, $\underline{h}_\tau$ = diameter of the largest ball which can be included in τ .

Let $h = \max_{\tau \in \mathcal{T}} h_\tau$ and $\rho^{-1} = C_0(\mathcal{T}) = \min_{\tau \in \mathcal{T}} \frac{h_\tau}{\underline{h}_\tau}$. If $\rho(\mathcal{T}) = \max_{\tau \in \mathcal{T}} \frac{h_\tau}{\underline{h}_\tau} \leq c$, we say $\mathcal{T}$ is shape regular.

Proposition 4.1

Assume triangle $\tau \in \mathcal{T}$, if its smallest angle is bounded below by a constant. Then $\mathcal{T}$ is shape-regular. 

T is an affine mapping of $\hat{\tau}$ onto τ just like Figure (4.1)

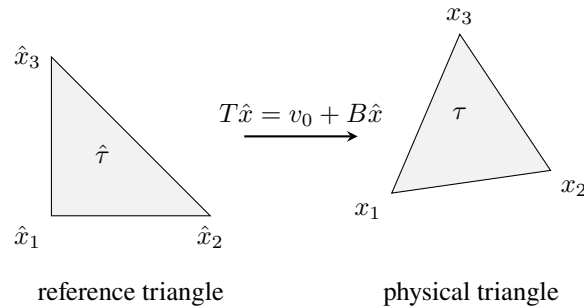


Figure 4.1: Affine mapping from the reference triangle $\hat{\tau}$ to a physical triangle τ

Lemma 4.3

$$|v|_{H^m(\tau)} \leq C |\det B|^{\frac{1}{2}} \cdot \|B^{-1}\|_2^m \cdot |\hat{v}|_{H^m(\hat{\tau})}$$

Here v is defined on τ and $\hat{v} = v \circ T$,

$$\|B\| = \sup_{x \in \mathbb{R}^d} \frac{|Bx|}{|x|} \quad (d = 2 \text{ for } \mathbb{R}^2)$$

Also,

$$|\hat{v}|_{H^m(\hat{\tau})} \leq C |\det B|^{-\frac{1}{2}} \|B\|_2^m |v|_{H^m(\tau)}$$



Proof We just proof the condition for H^1 . Idea: change of variable and chain rule.

Note that $v = \hat{v} \circ T^{-1}$ ($v(x) = \hat{v}(T^{-1}x)$)

$$T^{-1}x = \hat{x} + B^{-1}x$$

$$\nabla_x v = \nabla_{\hat{x}} \hat{v} \cdot D(T_x^{-1}) = \nabla_{\hat{x}} \hat{v} \cdot B^{-1}$$

$$\begin{aligned} \int_{\tau} |\nabla v|^2 dx &= \int_{\hat{\tau}} |\det B| \cdot |\nabla \hat{v} \cdot B^{-1}|^2 d\hat{x} \\ &= \int_{\hat{\tau}} |B^{-1} \nabla \hat{v}|^2 d\hat{x} \cdot |\det B| \\ &\leq |\det B| \cdot \|B^{-1}\|^2 \int_{\hat{\tau}} |\nabla \hat{v}|^2 d\hat{x} \end{aligned}$$

Thus

$$|v|_{H^1(\tau)}^2 \leq |\det B| \|B^{-1}\|_2^2 |\hat{v}|_{H^1(\hat{\tau})}^2$$

Here we note. $\|B\|_2 = \|B^T\|_2$, $\|B^{-1}\|_2 = \|B^{-T}\|_2$. (Exercise). and

$$\|B^{-T}w\| \leq \|B^{-T}\| \|w\| = \|B^{-1}\| |\omega|.$$

So

$$|v|_{H^1(\tau)} \leq \|B^{-1}\| \cdot |\det B|^{\frac{1}{2}} |\hat{v}|_{H^1(\hat{\tau})}.$$

We can thus similarly show the inequality for m .

Lemma 4.4

$$\|B\| \leq Ch_{\tau} \text{ and } \|B^{-1}\| \leq C_0 h_{\tau}^{-1} \quad (C_0 \text{ depends on } C_0(\tau)).$$

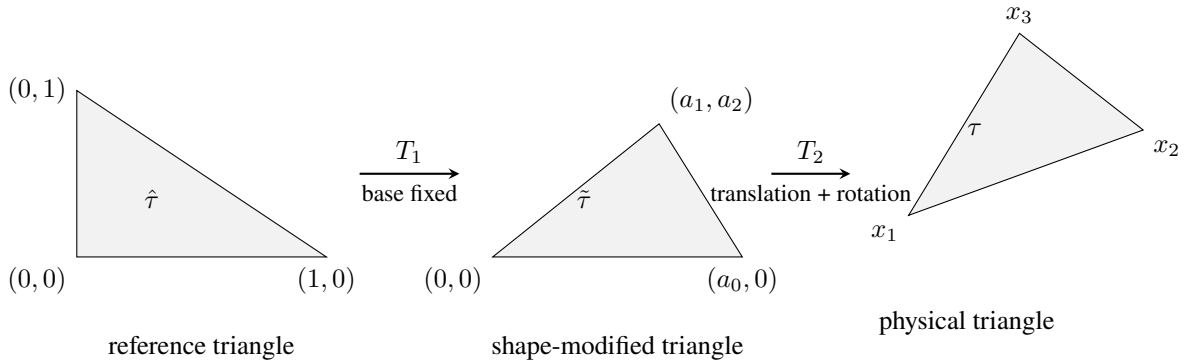


Figure 4.2: Two-step decomposition of the affine mapping

Proof $T_1 \hat{x} = B_1 \hat{x} = y$ where $B_1 = \begin{pmatrix} a_0 & a_1 \\ 0 & a_2 \end{pmatrix}$, $T_2 y = B_2 y + v_0$, thus $B = B_2 \circ B_1$.

Check B_2 is unitary i.e. $|B_2 x| = |x|$, $\|B_2\| = \|B_2^{-1}\| = 1$.

$$\|B\| = \|B_2 B_1\| \leq \|B_2\| \|B_1\| = \|B_1\|.$$

So $\|B_1\| \leq C \|B_1\|_\infty$ where $\|B_1\|_\infty = \max_{i,j=1,2} |(b_1)_{ij}|$. Since $a_0, a_1, a_2 \leq \text{diam } \tau = h_\tau$. We get

$$\|B\| \leq \|B_1\| \leq c \cdot h_\tau$$

On the otherhand

$$B_1^{-1} = \begin{pmatrix} a_0^{-1} & -a_1/a_2 \\ 0 & a_2^{-1} \end{pmatrix} = (a_0 a_2)^{-1} \begin{pmatrix} a_2 & -a_1 \\ a_0 & \end{pmatrix} \Rightarrow \|B_1^{-1}\| \leq C \cdot |(a_0 a_2)^{-1}| h_\tau$$

Note that $\text{Area}(\tau) = \frac{1}{2} a_0 a_2 \geq \pi \left(\frac{1}{2} \underline{h}_\tau\right)^2 \geq \frac{\pi}{4} c_0^2(\tau) \cdot h_\tau^2$. $\left(c_0(\tau) \leq \frac{h_\tau}{\underline{h}_\tau}\right)$.

$$\text{So } (a_0 a_2)^{-1} \leq c h_\tau^2 \Rightarrow \|B_1^{-1}\| \leq c h_\tau^{-1}$$

Here the constant c . depends $C_0^{-1}(\tau)$.

$$C_0^{-1}(\tau) = \max_{\tau \in \mathcal{J}} \frac{h_\tau}{\underline{h}_\tau}$$

if $c_0^{-1}(\tau) \leq c$, we say $\mathcal{J}$ is shape-regular.

Theorem 4.1 (Example of Sobolev embedaling)

For $k > \frac{d}{2}$, $H^k(\Omega) \subseteq C(\bar{\Omega})$ and there exists a constant $C > 0$ such that

$$\|u\|_{L^\infty(\Omega)} \leq C \|u\|_{H^k(\Omega)} \quad \text{for all } u \in H^k(\Omega).$$



Remark For Lagrange interpolant, we need function evaluation,

$$\{x_i\} \subset \mathcal{N}_k(\tau), \tau \in \mathcal{J}. \quad f(x_i) < +\infty \Rightarrow H^k(\Omega) \subseteq C(\bar{\Omega})$$

Theorem 4.2 (Interpolation error)

Let Ω be a polygonal domain. $\mathcal{J}$ is a shape regular triangulation of Ω (admissible).

$$S_h = \left\{ \phi \in C(\Omega) \text{ satisfying } \phi|_\tau \in \mathbb{P}^k \text{ for } \tau \in \mathcal{J} \right\}.$$

For $u \in H^m(\Omega)$, $k+1 \geq m > \frac{d}{2}$ and $l = 0, 1$.

$$\|u - \mathcal{I}_h u\|_{H^l(\Omega)} \leq c h^{m-l} \|u\|_{H^m(\Omega)}.$$

where $\mathcal{I}_h$ denotes the Lagrange finite element interpolation operator into S_h .



Proof

$$\int_\Omega |D^\alpha (u - \mathcal{I}_h u)|^2 dx = \sum_{\tau \in \mathcal{J}} \int_\tau |D^\alpha (u - \mathcal{I}_h u)|^2 dx$$

Consider

$$\begin{aligned} I_1 &= \int_\Omega |u - \mathcal{I}_h u|^2 dx = \sum_{\tau \in \mathcal{J}} \int_\tau |u - \mathcal{I}_h u|^2 dx \\ &= \sum_{\tau \in \mathcal{J}} \int_{\hat{\tau}} |\det B| \left| \widehat{u - \mathcal{I}_h u} \right|^2 d\hat{x} \quad (\hat{f} = f \circ T) \\ I_2 &= \int_\Omega |\nabla (u - \mathcal{I}_h u)|^2 dx \leq \sum_{\tau \in \mathcal{J}} |\det B| \|B^{-1}\|_2^2 \int_{\hat{\tau}} \left| \nabla \left(\widehat{u - \mathcal{I}_h u} \right) \right|^2 d\hat{x}. \end{aligned}$$

As T is linear, $\widehat{u - \mathcal{I}_h u} = \hat{u} - \widehat{\mathcal{I}_h u}$. Let $\mathcal{I}$ be the interpolation with respect to the nodes on $\hat{\tau}$, then

$$\widehat{\mathcal{I}_h u} = \mathcal{I} \hat{u} \quad (\text{Exercise})$$

So

$$I_2 \leq \sum_{\tau \in \mathcal{T}} |\det B| \|B^{-1}\|_2^2 \int_{\hat{\tau}} |\nabla(\hat{u} - \mathcal{I}\hat{u})|^2 d\hat{x}$$

$$I_1 = \sum_{\tau \in \mathcal{T}} |\det B| \int_{\hat{\tau}} |\hat{u} - \mathcal{I}\hat{u}|^2 d\hat{x}.$$

In fact $\mathcal{I}$ is stable in $H^m(\Omega)$.

$$\|\mathcal{I}\hat{w}\|_{H^m(\hat{\tau})} \leq C \|\hat{w}\|_{H^m(\hat{\tau})}.$$

Now we verify the stability of $\mathcal{I}$. Let $\hat{\varphi}_i$ be the finite element basis function for $\mathbb{P}^k$ on $\hat{\tau}$.

$$\mathcal{I}\hat{w} = \sum_{i=1}^{\frac{(k+1)(k+2)}{2}} w(\hat{x}_i) \hat{\varphi}_i(\hat{x}) \quad \hat{x}_i \in \mathcal{N}_{m-1}(\hat{\tau}) \quad \hat{\varphi}_i(\hat{x}_j) = \delta_{ij}.$$

$$\|\mathcal{I}\hat{w}\|_{H^m(\hat{\tau})} \leq \sum_{i=1}^{\dim(\mathbb{P}^k)} |\hat{w}(\hat{x}_i)| \|\hat{\varphi}_i\|_{H^m(\hat{\tau})}$$

$$\leq \underbrace{\max \|\hat{\varphi}_i\|_{H^m(\hat{\tau})} \cdot \dim(\mathbb{P}^k)}_{\text{constant}} \cdot c \|\hat{w}\|_{H^m(\hat{\tau})}$$

Note that $\mathcal{I}p = p$ for any $p \in \mathbb{P}^{m-1}$ ($m-1 \leq k$). Thus

$$|\hat{u} - \mathcal{I}\hat{u}| = |\hat{u} - p + \mathcal{I}p - \mathcal{I}\hat{u}|$$

$$\leq |\hat{u} - p| + |\mathcal{I}(\hat{u} - p)|.$$

$$\int_{\hat{\tau}} |\hat{u} - \mathcal{I}\hat{u}|^2 d\hat{x} \leq 2 \int_{\hat{\tau}} |\hat{u} - p|^2 d\hat{x} + 2 \int_{\hat{\tau}} |\mathcal{I}(\hat{u} - p)|^2 d\hat{x}$$

$$\leq C \int_{\hat{\tau}} |\hat{u} - p|^2 d\hat{x} \leq c \|\hat{u} - p\|_{H^m(\hat{\tau})}^2.$$

Similarly,

$$\int_{\hat{\tau}} |\nabla(\hat{u} - \mathcal{I}\hat{u})|^2 d\hat{x} \leq C \cdot \|\hat{u} - p\|_{H^m(\hat{\tau})}$$

Take the infimum over p and apply the Deny-Lions Lemma.

$$\int_{\hat{\tau}} |D^\alpha(\hat{u} - \mathcal{I}\hat{u})|^2 d\hat{x} \leq C \inf_{p \in \mathbb{P}^k} \|\hat{u} - p\|_{H^m(\hat{\tau})}^2 \leq C |\hat{u}|_{H^m(\hat{\tau})}^2 \quad (|\alpha| \leq 1).$$

Thus

$$I_1 \leq C \sum_{\tau \in \mathcal{T}} |\det B| |\hat{u}|_{H^m(\hat{\tau})}^2 \leq C \sum_{\tau \in \mathcal{T}} \|B\|^{2m} \cdot |u|_{H^m(\tau)}^2 \leq Ch^{2m} |u|_{H^m(\Omega)}^2$$

So

$$\|u - \mathcal{I}_h u\|_{L^2(\Omega)} \leq ch^m \|u\|_{H^m(\Omega)}$$

Then

$$I_2 \leq \sum_{\tau \in \mathcal{T}} C |\det B| \|B^{-1}\|_2^2 |\hat{u}|_{H^m(\hat{\tau})}^2 \leq \sum_{\tau \in \mathcal{T}} \|B^{-1}\|_2^2 \|B\|_2^{2m} |u|_{H^m(\tau)}^2$$

$$\leq C \sum_{\tau \in \mathcal{T}} \underline{h}_\tau^{-2} h_\tau^{2m} |u|_{H^m(\tau)}^2 \leq C \sum_{\tau \in \mathcal{T}} h_\tau^{2m-2} |u|_{H^m(\tau)}^2$$

$$\leq Ch^{2m-2} |u|_{H^m(\Omega)}^2$$

We get

$$|\nabla(u - \mathcal{I}_h u)|_{H^1(\Omega)} \leq ch^{m-1} \|u\|_{H^m(\Omega)}.$$

Summary.

$$\|u - \mathcal{I}_h u\|_{L^2(\Omega)} + h \|u - \mathcal{I}_h u\|_{H^1(\Omega)} \leq Ch^l \|u\|_{H^1(\Omega)} \quad 2 \leq l \leq k+1.$$

4.3 Inverse estimate

Lemma 4.5

Let τ be a triangle. p is a polynomial defined on τ . with $p \in \mathbb{P}^k$. Then

$$|p|_{H^1(\tau)} \leq Ch_\tau^{-1} \|p\|_{L^2(\tau)}.$$



Proof Since $\mathbb{P}^k$ is a finite dimensional space in τ . We know that any norm for $\mathbb{P}^k$ are equivalent. So for $\hat{p} \in \mathbb{P}_k(\hat{\tau})$

$$C_1 \|\hat{p}\|_{L^2(\hat{\tau})} \leq \|\hat{p}\|_{H^1(\hat{\tau})} \leq C_2 \|\hat{p}\|_{L^2(\hat{\tau})}$$

where C_1, C_2 , depends on k and $\hat{\tau}$. This implies that

$$|\hat{p}|_{H^1(\hat{\tau})} \leq C \|\hat{p}\|_{L^2(\hat{\tau})}.$$

By the previous Lemma, let $\hat{p} = p \circ T$ and we get

$$\begin{aligned} |p|_{H^1(\tau)}^2 &\leq |\det B| \|B^{-1}\|_2^2 |\hat{p}|_{H^1(\hat{\tau})}^2 \\ &\leq C_2 |\det B| \|B^{-1}\|_2^2 \|\hat{p}\|_{L^2(\hat{\tau})}^2 \\ &\leq C_2 |\det B| \|B^{-1}\|_2^2 |\det B|^{-1} \|p\|_{L^2(\tau)}^2 \\ &= C_2 \|B^{-1}\|_2^2 \|p\|_{L^2(\tau)}^2 \end{aligned}$$

Therefore, since $\mathcal{J}$ is shape regular, we have

$$|p|_{H^1(\tau)} \leq C \|B^{-1}\|_2 \|p\|_{L^2(\tau)} \leq ch_\tau^{-1} \|p\|_{L^2(\tau)}$$

Remark This also implies that

$$\|p\|_{H^1(\tau)} \leq ch_\tau^{-1} \|p\|_{L^2(\tau)}$$

It is trivial to check that

$$\|p\|_{H^1(\tau)}^2 = \|p\|_{L^2(\tau)}^2 + |p|_{H^1(\tau)}^2 \leq \|p\|_{L^2(\tau)}^2 + C^2 h_\tau^{-2} \|p\|_{L^2(\tau)}^2 \leq Ch_\tau^{-2} \|p\|_{L^2(\tau)}^2,$$

Define $h = \max_{\tau \in \mathcal{J}} h_\tau$. If for all $\tau \in \mathcal{J}$, $\min_{\tau} h_\tau \geq ch$, we say $\mathcal{J}$ is quasi-uniform. It is equivalent to $\exists c_1, c_2 > 0, \forall \tau \in \mathcal{J}, c_1 h \leq h_\tau \leq c_2 h$.

We can further show that for $p \in S_h$.

$$\begin{aligned} |p|_{H^1(\Omega)}^2 &\leq \|p\|_{H^1(\Omega)}^2 = \sum_{\tau \in \mathcal{J}} \|p\|_{H^1(\tau)}^2 \leq C \sum_{\tau \in \mathcal{J}} h_\tau^{-2} \|p\|_{L^2(\tau)}^2 \\ &\leq c \sum_{\tau \in \mathcal{J}} h^{-2} \|p\|_{L^2(\tau)}^2 = ch^{-2} \|p\|_{L^2(\Omega)}^2 \end{aligned}$$

i.e.

$$\|p\|_{H^1(\Omega)} \leq Ch^{-1} \|p\|_{L^2(\Omega)}, \quad \forall p \in S_h$$

This is the global inverse inequality.

Chapter 5 high dimensional problem

5.1 Two-Dimensional Problem

Consider the following boundary value problem:

$$\begin{cases} u - \Delta u = f & \text{in } \Omega \\ \frac{\partial u}{\partial \mathbf{n}} = \nabla u \cdot \mathbf{n} = g & \text{on } \partial\Omega \end{cases} \quad (5.1)$$

Multiply (5.1) by a test function $\varphi \in C^\infty(\bar{\Omega})$:

$$\int_{\Omega} u\varphi - \int_{\Omega} \Delta u\varphi = \int_{\Omega} f\varphi.$$

Integrate by parts (Green's formula):

$$- \int_{\Omega} \Delta u \cdot \varphi = - \int_{\partial\Omega} (\nabla u \cdot \mathbf{n})\varphi \, ds + \int_{\Omega} \nabla u \cdot \nabla \varphi \, dx.$$

Since $\nabla u \cdot \mathbf{n} = g$ on $\partial\Omega$, substituting this back gives:

$$\int_{\Omega} u\varphi \, dx + \int_{\Omega} \nabla u \cdot \nabla \varphi \, dx - \int_{\partial\Omega} g\varphi \, ds = \int_{\Omega} f\varphi \, dx.$$

Rearranging the terms:

$$\underbrace{\int_{\Omega} (u\varphi + \nabla u \cdot \nabla \varphi) \, dx}_{A(u,\varphi)} = \underbrace{\int_{\Omega} f\varphi \, dx + \int_{\partial\Omega} g\varphi \, ds}_{\langle F, \varphi \rangle}.$$

Note that $A(u, \varphi) = (u, \varphi)_{H^1(\Omega)}$, which is the standard inner product in H^1 .

So our weak formulation is: find $u \in H^1(\Omega)$ satisfying

$$A(u, \varphi) = \langle F, \varphi \rangle, \quad \forall \varphi \in H^1(\Omega).$$

Definition 5.1

A linear map $F : H \rightarrow \mathbb{R}$ on a Hilbert space H is called a linear functional. It is bounded (or continuous) if there exists a constant $C \geq 0$ satisfying

$$|\langle F, \varphi \rangle| \leq C \|\varphi\|_H, \quad \forall \varphi \in H.$$



Theorem 5.1 (Riesz Representation Theorem)

Let H be a Hilbert space and F be a bounded linear functional on H . Then there exists a unique element $u \in H$ such that

$$(u, \varphi)_H = \langle F, \varphi \rangle, \quad \forall \varphi \in H.$$

Moreover, $\|u\|_H = \|F\|_{H'}$.



Since $A(u, \varphi) = (u, \varphi)_{H^1}$ is exactly the inner product, we have shown that (5.1) admits a unique weak solution.

5.1.1 Generalization

Consider the general elliptic operator:

$$Lu = - \sum_{i,j=1}^d \frac{\partial}{\partial x_i} \left(a_{ij}(x) \frac{\partial}{\partial x_j} u \right) + q(x)u. \quad (5.2)$$

Assumptions:

1. $a_{ij} \in L^\infty(\Omega)$ and satisfy the **Uniform Ellipticity Condition**: There exists a constant $\nu > 0$ such that

$$\sum_{i,j=1}^d a_{ij}(x) \xi_i \xi_j \geq \nu |\xi|^2, \quad \forall \xi \in \mathbb{R}^d, \text{ a.e. } x \in \Omega.$$

(Note: Do not confuse the vector $\xi \in \mathbb{R}^d$ with the test function v).

2. $q \in L^\infty(\Omega)$ and there is a constant $q_0 > 0$ such that $q(x) \geq q_0$ a.e. in Ω .

Define the co-normal derivative on $\partial\Omega$:

$$\frac{\partial u}{\partial \nu_A} = \sum_{i,j=1}^d n_i a_{ij} \frac{\partial u}{\partial x_j} = (\mathbf{n}^T A \nabla u).$$

We want to solve the Neumann problem:

$$\begin{cases} Lu = f & \text{in } \Omega, \\ \frac{\partial u}{\partial \nu_A} = g & \text{on } \partial\Omega. \end{cases} \quad (5.3)$$

(Given $f \in L^2(\Omega), g \in L^2(\partial\Omega)$).

Derivation: Take $\varphi \in C^\infty(\bar{\Omega})$.

$$\int_{\Omega} Lu \varphi \, dx = \int_{\Omega} f \varphi \, dx.$$

Analyze the Left Hand Side (LHS):

$$\begin{aligned} \text{LHS} &= - \sum_{i,j=1}^d \int_{\Omega} \frac{\partial}{\partial x_i} \left(a_{ij}(x) \frac{\partial u}{\partial x_j} \right) \varphi \, dx + \int_{\Omega} qu \varphi \, dx \\ &= \sum_{i,j=1}^d \int_{\Omega} a_{ij}(x) \frac{\partial u}{\partial x_j} \frac{\partial \varphi}{\partial x_i} \, dx - \sum_{i,j=1}^d \int_{\partial\Omega} n_i a_{ij}(x) \frac{\partial u}{\partial x_j} \varphi \, ds + \int_{\Omega} qu \varphi \, dx \\ &= \sum_{i,j=1}^d \int_{\Omega} a_{ij} \frac{\partial u}{\partial x_j} \frac{\partial \varphi}{\partial x_i} \, dx - \int_{\partial\Omega} \underbrace{\left(\sum_{i,j=1}^d n_i a_{ij} \frac{\partial u}{\partial x_j} \right)}_{\frac{\partial u}{\partial \nu_A} = g} \varphi \, ds + \int_{\Omega} qu \varphi \, dx. \end{aligned}$$

Variational formulation: Find $u \in H^1(\Omega)$ such that:

$$A(u, \varphi) = \langle F, \varphi \rangle, \quad \forall \varphi \in H^1(\Omega).$$

Here:

$$\begin{aligned} A(u, \varphi) &= \int_{\Omega} qu \varphi \, dx + \sum_{i,j=1}^d \int_{\Omega} a_{ij} \frac{\partial u}{\partial x_j} \frac{\partial \varphi}{\partial x_i} \, dx, \\ \langle F, \varphi \rangle &= \int_{\Omega} f \varphi \, dx + \int_{\partial\Omega} g \varphi \, ds. \end{aligned}$$

5.1.2 Properties of the Bilinear Form

1. **Boundedness (Continuity):** A is bounded on $H^1(\Omega) \times H^1(\Omega)$.

$$\begin{aligned} \left| \int_{\Omega} q uv \right| &\leq \|q\|_{L^\infty} \|u\|_{L^2} \|v\|_{L^2} \leq \|q\|_{L^\infty} \|u\|_{H^1} \|v\|_{H^1}. \\ \left| \sum_{i,j} \int_{\Omega} a_{ij} \frac{\partial u}{\partial x_j} \frac{\partial v}{\partial x_i} \right| &\leq \sum_{i,j} \|a_{ij}\|_{L^\infty} \left\| \frac{\partial u}{\partial x_j} \right\|_{L^2} \left\| \frac{\partial v}{\partial x_i} \right\|_{L^2} \\ &\leq C \sum_{i,j} \|\nabla u\|_{L^2} \|\nabla v\|_{L^2} \leq C' \|u\|_{H^1} \|v\|_{H^1}. \end{aligned}$$

Thus, $|A(u, v)| \leq M \|u\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)}$.

2. **Boundedness of Functional F :** By Cauchy-Schwarz and Trace Theorem:

$$|\langle F, \varphi \rangle| \leq \|f\|_{L^2} \|\varphi\|_{L^2} + \|g\|_{L^2(\partial\Omega)} \|\varphi\|_{L^2(\partial\Omega)} \leq C \|\varphi\|_{H^1(\Omega)}.$$

3. **Coercivity:** We verify that $A(u, u) \geq \alpha \|u\|_{H^1(\Omega)}^2$.

$$\begin{aligned} A(u, u) &= \int_{\Omega} q u^2 dx + \int_{\Omega} \sum_{i,j=1}^d a_{ij} \frac{\partial u}{\partial x_i} \frac{\partial u}{\partial x_j} dx \\ &\geq q_0 \|u\|_{L^2}^2 + \nu \|\nabla u\|_{L^2}^2 \quad (\text{using ellipticity assumption}) \\ &\geq \min(q_0, \nu) (\|u\|_{L^2}^2 + \|\nabla u\|_{L^2}^2) \\ &= \alpha \|u\|_{H^1(\Omega)}^2. \end{aligned}$$

5.1.3 Symmetry and Existence

Assume that the coefficient matrix is symmetric, i.e., $a_{ij}(x) = a_{ji}(x)$ for $i, j = 1, \dots, d$. Check symmetry of A :

$$\begin{aligned} A(v, u) &= \int_{\Omega} q v u + \sum_{i,j=1}^d \int_{\Omega} a_{ij} \frac{\partial v}{\partial x_j} \frac{\partial u}{\partial x_i} \\ &\quad \text{Let } k = j, l = i \text{ (rename indices)} \\ &= \int_{\Omega} q u v + \sum_{l,k=1}^d \int_{\Omega} a_{lk} \frac{\partial v}{\partial x_k} \frac{\partial u}{\partial x_l} \\ &\quad \text{Since } a_{lk} = a_{kl} \text{ (symmetry assumption)} \\ &= \int_{\Omega} q u v + \sum_{k,l=1}^d \int_{\Omega} a_{kl} \frac{\partial u}{\partial x_l} \frac{\partial v}{\partial x_k} = A(u, v). \end{aligned}$$

Conclusion: Since $A(\cdot, \cdot)$ is a **symmetric**, **bounded**, and **coercive** bilinear form, it defines an inner product on $H^1(\Omega)$ equivalent to the standard one. By the Riesz Representation Theorem applied to the Hilbert space $(H^1(\Omega), A(\cdot, \cdot))$, there exists a unique solution $u \in H^1(\Omega)$ to the problem.

Remark If $a_{ij} \neq a_{ji}$, A is not symmetric. We cannot use Riesz directly. Instead, we must use the **Lax-Milgram Theorem**, which only requires boundedness and coercivity (not symmetry).

5.2 Lax-Milgram

Theorem 5.2 (Lax-Milgram)

Let V be a Hilbert space. Let $A : V \times V \rightarrow \mathbb{R}$ be a bounded, coercive bilinear form, and $F : V \rightarrow \mathbb{R}$ be a bounded linear functional. Then there exists a unique $u \in V$ satisfying

$$A(u, \varphi) = F(\varphi), \quad \forall \varphi \in V.$$



Remark We do not need to assume the symmetry of the coefficients, i.e., $a_{ij}(x) = a_{ji}(x)$ is not required. This is the main difference from the Riesz Representation Theorem.

Example 5.1 Consider the following boundary value problem:

$$\begin{cases} u - \Delta u + \alpha u_x = f & \text{in } \Omega, \\ \frac{\partial u}{\partial \mathbf{n}} = 0 & \text{on } \partial\Omega. \end{cases}$$

Multiplying by a test function $\varphi \in H^1(\Omega)$ and integrating by parts, we obtain the variational formulation. Note that the boundary term vanishes due to the Neumann boundary condition:

$$\begin{aligned} \int_{\Omega} (u - \Delta u + \alpha u_x) \varphi &= \int_{\Omega} u \varphi + \int_{\Omega} \nabla u \cdot \nabla \varphi - \int_{\partial\Omega} \frac{\partial u}{\partial \mathbf{n}} \varphi + \alpha \int_{\Omega} u_x \varphi \\ &= \int_{\Omega} u \varphi + \int_{\Omega} \nabla u \cdot \nabla \varphi + \alpha \int_{\Omega} u_x \varphi. \end{aligned}$$

Define the bilinear form $A(\cdot, \cdot)$ and linear functional $F(\cdot)$ as:

$$A(u, \varphi) = \int_{\Omega} u \varphi + \int_{\Omega} \nabla u \cdot \nabla \varphi + \alpha \int_{\Omega} u_x \varphi, \quad F(\varphi) = \int_{\Omega} f \varphi.$$

1. Boundedness: Using the Cauchy-Schwarz inequality:

$$\left| \int_{\Omega} u_x \varphi \right| \leq \|u_x\|_{L^2} \|\varphi\|_{L^2} \leq \|u\|_{H^1} \|\varphi\|_{H^1}.$$

It is easy to check that other terms are also bounded, so A is bounded on $H^1(\Omega) \times H^1(\Omega)$.

2. Coercivity: We need to estimate the convection term $\int_{\Omega} u_x u$.

Method 1 (Young's Inequality):

$$\begin{aligned} \left| \int_{\Omega} u_x u \right| &\leq \|u_x\|_{L^2} \|u\|_{L^2} \leq \frac{1}{2} \|u_x\|_{L^2}^2 + \frac{1}{2} \|u\|_{L^2}^2 \\ &\leq \frac{1}{2} (\|\nabla u\|_{L^2}^2 + \|u\|_{L^2}^2) = \frac{1}{2} \|u\|_{H^1}^2. \end{aligned}$$

Thus,

$$\begin{aligned} A(u, u) &= \|u\|_{L^2}^2 + \|\nabla u\|_{L^2}^2 + \alpha \int_{\Omega} u_x u \\ &\geq \|u\|_{H^1}^2 - |\alpha| \left| \int_{\Omega} u_x u \right| \\ &\geq \|u\|_{H^1}^2 - \frac{|\alpha|}{2} \|u\|_{H^1}^2 = \left(1 - \frac{|\alpha|}{2}\right) \|u\|_{H^1}^2. \end{aligned}$$

Therefore, if $|\alpha| < 2$, the bilinear form is coercive.

Method 2 (Integration by parts / Trace Theorem): Alternatively, use the divergence theorem noting that $uu_x = \frac{1}{2} \partial_x(u^2)$:

$$\begin{aligned} \int_{\Omega} u_x u \, dx &= \frac{1}{2} \int_{\Omega} \frac{\partial}{\partial x} (u^2) \, dx \\ &= \frac{1}{2} \int_{\partial\Omega} u^2 n_x \, ds \quad (\text{where } n_x \text{ is the x-component of } \mathbf{n}). \end{aligned}$$

Since $|n_x| \leq 1$, we have:

$$\left| \int_{\Omega} u_x u \right| \leq \frac{1}{2} \int_{\partial\Omega} u^2 ds \leq \frac{C_{tr}^2}{2} \|u\|_{H^1(\Omega)}^2,$$

where C_{tr} is the constant from the trace inequality $\|v\|_{L^2(\partial\Omega)} \leq C_{tr} \|v\|_{H^1(\Omega)}$.

This leads to:

$$A(u, u) \geq \left(1 - \frac{|\alpha| C_{tr}^2}{2}\right) \|u\|_{H^1}^2.$$

This provides coercivity if α is sufficiently small relative to the domain geometry (which determines C_{tr}).

5.3 inf-sup condition

Ladyzhenskaya-Babuška-Brezzi condition.

5.3.1 Matrix Case (On a finite dimensional case)

1. All norm on finite dimensional space are equivalent.

Let space V (finite dimensional) has two norms $\|\cdot\|_i, i = 1, 2$. Then there are constants C_1, C_2 , such that ($C_1 > 0$)

$$C_1 \|u\|_2 \leq \|u\|_1 \leq C_2 \|u\|_2.$$

2. All linear operators are bounded.

3. All bilinear forms are bounded.

Consider $M : \mathbb{R}^m \rightarrow \mathbb{R}^n$. M is a $n \times m$ matrix.

$$A(u, v) = (Mu) \cdot v, \quad u \in \mathbb{R}^m, v \in \mathbb{R}^n.$$

4. All linear functionals are bounded.

Given a linear functional F on $\mathbb{R}^n$, we seek the solution $u \in \mathbb{R}^m$ satisfying $A(u, \varphi) = \langle F, \varphi \rangle, \quad \forall \varphi \in \mathbb{R}^n$.

This is the same as

$$Mu = b, \quad b_i = \langle F, e_i \rangle.$$

For uniqueness, we need

$$M : \mathbb{R}^m \xrightarrow{1:1} \mathbb{R}^n$$

$$M : \mathbb{R}^m \xrightarrow{1:\text{onto}} \text{Range}(M) \subseteq \mathbb{R}^n$$

$$\exists M^{-1} = \text{Range } M \mapsto \mathbb{R}^m.$$

M^{-1} is linear and hence bounded. There exists a C satisfying

$$|M^{-1}y| \leq C|y|, \quad y \in \text{Range } M.$$

This can be rewritten as $x = M^{-1}y$

$$|x| \leq C|y| \text{ for all } x \in \mathbb{R}^m$$

$$|y| = \sup_{\substack{y \in \mathbb{R}^n \\ y \neq 0}} \frac{(Mx, y)}{|y|}$$

So.

$$|x| \leq C \sup_{y \in \mathbb{R}^n} \frac{(Mx, y)}{|y|}, \quad \forall x \in \mathbb{R}^m$$

This is called the inf-sup condition. The minimal C satisfies

$$\frac{1}{c} = \inf_{x \in \mathbb{R}^m} \sup_{y \in \mathbb{R}^n} \frac{(Mx, y)}{|x||y|}$$

We are looking for solution to $Mu = b$. This has a solution only if $b \in \text{Range}(M)$.

$$\mathbb{R}^n = \text{Range } M \oplus \text{Ker } M^\top \quad (\text{linear algebra}).$$

$b \in \text{Range } M$ iff $b \perp \text{Ker } M^\top$, i.e. $b \in \{y \in \mathbb{R}^n, (y, \xi) = 0, \forall \xi \in \text{Ker } M^\top\}$.

$\xi \in \text{Ker } M^\top$ iff $M^\top \xi = 0$, iff $(M^\top \xi, x) = 0$ for all $x \in \mathbb{R}^m$ iff $(\xi, Mx) = 0 \quad \forall x \in \mathbb{R}^m$.

Compatibility condition $Mu = b$ has a solution iff $(b, \xi) = 0$, for all $\xi \in W := \{y \in \mathbb{R}^n \text{ and that } A(x, y) = 0 \quad \forall x \in \mathbb{R}^n\}$.

5.3.2 LBB condition: Extension to infinite dimensional case

Assumption

1. U, V are Banach spaces;
2. $A(u, v)$ is a bounded bilinear form on $U \times V$, namely

$$|A(u, v)| \leq c \|u\|_U \|v\|_V, u \in U, v \in V$$

3. Inf-sup condition, $\exists \beta > 0$. satisfying

$$\|u\|_U \leq \beta \cdot \sup_{v \in V} \frac{A(u, v)}{\|v\|_V}, \text{ for all } u \in U.$$

4. F is a bounded linear functional on V .
5. F satisfies the compactibility condition. iff $\langle F, \xi \rangle = 0$. for all $\xi \in W := \{\xi \in V, A(u, \xi) = 0 \text{ for all } u \in U\} \subseteq V$ closed subspace.

Theorem 5.3

Assume that $(A_1) - (A_4)$ hold and let V is reflexive. Then $\exists$ unique $u \in U$ satisfying

$$A(u, \varphi) = \langle F, \varphi \rangle, \quad \forall \varphi \in V.$$

iff F satisfies the compactibility condition.



Remark All Hilbert spaces are reflexive.

Lax-Milgram Theorem: consider $U = V = \text{a Hilbert space}$. Assume $A(\cdot, \cdot)$ and F satisfy $(A_2), (A_4)$, respectively and that A is coercive on $U \times U$ i.e.

$$\|u\|^2 \leq \alpha A(u, u). \quad \forall u \in U.$$

Then $\exists$ unique $u \in U$ satisfying. $A(u, \varphi) = F(\varphi), \quad \forall \varphi \in U$.

Proof of the Remark: To apply the previous theorem we need only check (A_3) and compactibility condition.

Compactibility condition. If $w \in W$ then $A(u, w) = 0 \quad \forall u \in U$.

$$\Rightarrow \text{Since } w \in U \Rightarrow 0 = \alpha \cdot A(w, w) = \|w\|_U^2 \Rightarrow w = 0.$$

$$\Rightarrow \langle F, w \rangle = 0 \quad \forall w \in W \quad (\text{since } w = 0).$$

For (A_3) , the coercivity implies that

$$\|u\|_v \leq \alpha \cdot \frac{A(u, u)}{\|u\|_U} \leq \alpha \sup \frac{A(u, v)}{\|v\|_V}.$$

5.3.2.1 Reflexivity

Let W be a Banach space, $W^* = \{ \text{set of bounded linear functional} \}$.

Given $F \in W^*$, we define $\|F\|_{W^*}$ to be the minimal constant $C > 0$ satisfying

$$|\langle F, \varphi \rangle| \leq c \|\varphi\|_W, \quad \forall \varphi \in W$$

So W^* is a Banach space under this norm.

Let $W^{**} = \{ \text{set of bounded linear functionals on } W^* \}$ (also a Banach space).

We identify $w \in W$ with $F_w \in W^{**}$ satisfying $\langle F_w, G \rangle = \langle G, w \rangle$. So $W \subseteq W^{**}$.

Definition 5.2

W is reflexive if $W = W^{**}$.



Remark Hilbert spaces are reflexive by the Riesz Representation Theorem.

$L^p(\Omega)$ is reflexive for $p \in (1, \infty)$. $\Rightarrow W^{k,p}(\Omega)$ is reflexive.

Theorem 5.4 (Hahn-Banach theorem)

If F is a bounded linear functional on a subspace W_0 of normed linear space W . Then F can be extended to $\tilde{F}$ on W with

$$\|F\|_{W_0} = \|\tilde{F}\|_W.$$



Theorem 5.5 (Open mapping theorem)

If a linear bounded. $A : U \xrightarrow{1:1 \text{ onto}} V$ (U, V are Banach spaces). Then A^{-1} is bounded i.e. $\exists C$ with

$$\|A^{-1}f\|_U \leq C\|f\|_V \quad \forall f \in V.$$



Proof Proof of the theorem:

Define $\tilde{A} : U \rightarrow V^*$ by.

$$\langle \tilde{A}\theta, v \rangle = A(\theta, v). \quad \forall \theta \in U, v \in V$$

Inf- sup condition.

$$\|\theta\|_U \leq \alpha \sup_{v \in V} \frac{\langle \tilde{A}\theta, v \rangle}{\|v\|_V} \quad (5.4)$$

Check Range $(\tilde{A})$ is closed. Let $F_n \rightarrow F$ in Range $(\tilde{A})$.

Let $\tilde{A}\theta_n = F_n$. By (5.4) $\theta_n \rightarrow \theta$ in U and it is easy to check that $\tilde{A}\theta = F$.

So Range $(\tilde{A})$ is closed. (5.4) also implies that $\tilde{A}$ is one-to-one.

Clearly by the open mapping theorem, $\tilde{A}u = F \in V^*$ has a solution iff $F \in \text{Range } \tilde{A}$ in which case. the solution is unique.

Claim: $F \in \text{Range } \tilde{A}$ if $\Leftrightarrow \langle F, \varphi \rangle = 0, \quad \forall \varphi \in W$.

If $F \in \text{Range } \tilde{A}$, then $F = \tilde{A}\theta$ for some $\theta \in U$.

For $\varphi \in W, \langle F, \varphi \rangle = \langle \tilde{A}\theta, \varphi \rangle = A(\theta, \varphi) = 0$. (Since $\varphi \in W$).

Conversely $F \notin \text{Range } \tilde{A}$, define ℓ on Range $(\tilde{A}) \oplus \text{span } F$ by.

$$\ell(\omega^* + \beta F) = \beta. \quad \omega^* \in \text{Range } \tilde{A}.$$

Since Range $(\tilde{A})$ is closed and $F \notin \text{Range } \tilde{A}$, $\exists \varepsilon > 0$, such that

$$\|F - w^*\|_{V^*} > \varepsilon \quad \text{for all } w^* \in \text{Range } \tilde{A}$$

If $p \neq 0$.

$$\|w^* - \beta F\|_{V^*} = |\beta| \cdot \left\| \frac{1}{\beta} w^* - F \right\|_{V^*} > \varepsilon |\beta|$$

$$|\beta| = |\ell^*(\omega^* + \beta F)| \leq \frac{1}{\varepsilon} \|\omega^* + \beta F\|_{V^*} \Rightarrow \ell \text{ is bounded.}$$

By the Hahn-Banach theorem, we can extend ℓ to a bounded linear functional $\tilde{\ell}$ on V^* . Since V is reflexive, there is a $v \in V$ satisfying

$$\langle \hat{\ell}, w^* \rangle = \langle w^*, v \rangle. \quad \forall w^* \in V^*.$$

Consider $w^* = F$.

$$\langle F, v \rangle = \langle \tilde{\ell}, F \rangle = 1.$$

Also, for $u \in U$,

$$0 = \ell(\tilde{A}u) = \tilde{\ell}(\tilde{A}u) = \langle \tilde{A}u, v \rangle = A(u, v) \Rightarrow v \in W$$

i.e. F does not satisfy the compactibility condition.

Since the problem: Find $u \in U$ satisfying $A(u, \varphi) = F(\varphi)$ is the same as $\tilde{A}u = F$. Our proof is complete.

Remark Moreover

$$\|u\|_U \leq \sup_{\varphi \in V} \frac{A(u, \varphi)}{\|\varphi\|_V} = \alpha \cdot \sup_{\varphi \in V} \frac{\langle F, \varphi \rangle}{\|\varphi\|_V} = \alpha \cdot \|F\|_{V^*}$$

5.4 Galerkin finite element approximation

5.4.1 Variational Formulation

Assume we have a variational formulation which satisfies the properties of Lax-Milgram:

1. V is a Hilbert space.
2. $A(\cdot, \cdot)$ is a bounded bilinear form on V . That is, there exists a constant $\|A\|$ such that:

$$|A(u, v)| \leq \|A\| \|u\|_V \|v\|_V, \quad \forall u, v \in V.$$

3. F is a bounded functional on V .
4. A is coercive, i.e., there exists $\alpha > 0$ satisfying:

$$\alpha \|v\|_V^2 \leq A(v, v), \quad \forall v \in V.$$

Let $V_h \subseteq V$, then V_h satisfies (1)-(4) and hence there is a unique solution to: Find $u_h \in V_h$ s.t.

$$A(u_h, \varphi) = \langle F, \varphi \rangle, \quad \forall \varphi \in V_h.$$

So u_h is the Galerkin approximation to the solution $u \in V$ of:

$$A(u, \varphi) = \langle F, \varphi \rangle, \quad \forall \varphi \in V.$$

Subtracting the above equation gives (Galerkin orthogonality):

$$A(u - u_h, \varphi) = 0, \quad \forall \varphi \in V_h. \tag{5.5}$$

5.4.2 Céa's Lemma

Lemma 5.1 (Céa's Lemma)

$$\|u - u_h\|_V \leq \frac{1}{\alpha} \|A\| \inf_{v_h \in V_h} \|u - v_h\|_V. \quad (5.6)$$

Here $\|A\|$ is the smallest constant satisfying

$$|A(\omega, v)| \leq \|A\| \|\omega\|_V \|v\|_V.$$

This is a quasi-optimal estimate.



Proof By coercivity:

$$\begin{aligned} \|u - u_h\|_V^2 &\leq \frac{1}{\alpha} A(u - u_h, u - u_h) \\ &= \frac{1}{\alpha} A(u - u_h, u - v_h + v_h - u_h) \quad (\text{for any } v_h \in V_h) \\ &= \frac{1}{\alpha} A(u - u_h, u - v_h) \quad (\text{since } A(u - u_h, v_h - u_h) = 0) \\ &\leq \frac{1}{\alpha} \|A\| \|u - u_h\|_V \|u - v_h\|_V. \end{aligned}$$

Dividing by $\|u - u_h\|_V$, we get the result.

$$\|u - u_h\|_V \leq \frac{1}{\alpha} \|A\| \|u - v_h\|_V \quad \forall v_h \in V_h \implies \|u - u_h\|_V \leq \frac{\|A\|}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_V$$

5.4.3 L^2 Error Estimates (Nitsche's Trick/Duality)

Now we let $V_h = S_h = \{v_h \in H^1 : v_h|_\tau \in \mathbb{P}^k\}$. According to our finite element interpolation error estimates, we have:

$$\|u - u_h\|_V \leq C \|u - \mathcal{I}_h u\|_V \leq C \|u - \mathcal{I}_h u\|_{H^1} \leq C h^{\ell-1} \|u\|_{H^\ell(\Omega)},$$

where $2 \leq \ell \leq k+1$.

Assumption: Let Ω be convex and $w \in V \subseteq H^1(\Omega)$ be the unique solution to the dual problem:

$$A(\theta, w) = (\theta, g), \quad \forall \theta \in V.$$

Under the assumptions of Lax-Milgram, w is well defined, thus $w \in H^2(\Omega)$. We assume full elliptic regularity:

$$\|w\|_{H^2(\Omega)} \leq C \|g\|_{L^2(\Omega)}, \quad \forall g \in L^2(\Omega).$$

Theorem 5.6 (Nitsche's trick / Duality)

Assume the setup for Lax-Milgram, $V \subseteq H^1(\Omega)$, and full elliptic regularity. Then:

$$\|u - u_h\|_{L^2(\Omega)} \leq C h^\ell \|u\|_{H^\ell(\Omega)}.$$



Proof Let $e_h = u - u_h$. Let $w \in V$ solve:

$$A(\theta, w) = (\theta, e_h), \quad \forall \theta \in V.$$

So $e_h = u - u_h \in V \subset H^1(\Omega) \equiv L^2(\Omega)$. Set $\theta = e_h$ to get:

$$\begin{aligned} \|e_h\|_{L^2(\Omega)}^2 &= (e_h, e_h) = A(e_h, w) \\ &= A(e_h, w - \chi), \quad \forall \chi \in S_h \quad (\text{by Galerkin orthogonality}). \end{aligned}$$

Thus,

$$\|e_h\|_{L^2(\Omega)}^2 \leq \|A\| \|e_h\|_{H^1(\Omega)} \|w - \chi\|_{H^1(\Omega)}.$$

Taking $\chi = \mathcal{I}_h w$ gives:

$$\|w - \mathcal{I}_h w\|_{H^1(\Omega)} \leq Ch \|w\|_{H^2(\Omega)} \leq Ch \|e_h\|_{L^2(\Omega)}.$$

Combining the above estimates:

$$\|e_h\|_{L^2(\Omega)}^2 \leq \|A\| \|e_h\|_{H^1} \cdot Ch \|e_h\|_{L^2(\Omega)}.$$

Therefore:

$$\|e_h\|_{L^2(\Omega)} \leq Ch \|e_h\|_{H^1(\Omega)}.$$

Finally, using the energy norm estimate:

$$\|u - u_h\|_{L^2(\Omega)} \leq Ch \cdot Ch^{\ell-1} \|u\|_{H^\ell} = Ch^\ell \|u\|_{H^\ell}, \quad 2 \leq \ell \leq k+1$$

5.5 Poincare Inequality

Consider the problem:

$$\begin{aligned} -\Delta u &= f \quad \text{in } \Omega = (0,1)^2 \\ u(x) &= 0 \quad \text{on } \Gamma_D \text{ (essential BC, imposed strongly in } V) \\ \frac{\partial u}{\partial \mathbf{n}}(x) &= g \quad \text{on } \Gamma_N \text{ (natural BC, imposed weakly)} \end{aligned} \tag{5.7}$$

Derivation of the weak form:

$$\begin{aligned} \int_{\Omega} f \varphi &= \int_{\Omega} -\Delta u \varphi \\ &= - \int_{\partial\Omega} \frac{\partial u}{\partial \mathbf{n}} \varphi + \int_{\Omega} \nabla u \cdot \nabla \varphi \\ &= - \int_{\Gamma_N} \underbrace{\frac{\partial u}{\partial \mathbf{n}}}_{=g} \varphi - \int_{\Gamma_D} \underbrace{\frac{\partial u}{\partial \mathbf{n}}}_{=0} \varphi + \int_{\Omega} \nabla u \cdot \nabla \varphi \\ &= - \int_{\Gamma_N} g \varphi + \int_{\Omega} \nabla u \cdot \nabla \varphi \end{aligned}$$

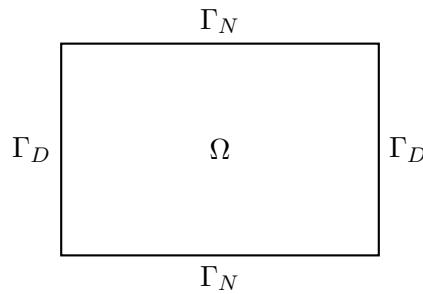


Figure 5.1: Domain Ω with mixed boundary conditions.

Define the Test Space (where we take test functions):

$$V = \{\varphi \in H^1(\Omega) : \varphi = 0 \text{ on } \Gamma_D\}.$$

Since the trace operator is continuous, i.e., $\|v\|_{L^2(\Gamma_D)} \leq C \|v\|_{H^1(\Omega)}$, V is a closed subspace of $H^1(\Omega)$.

Weak formulation: Find $u \in V$ satisfying

$$A(u, \varphi) = L(\varphi), \quad \forall \varphi \in V.$$

Here $A(u, \varphi) = \int_{\Omega} \nabla u \cdot \nabla \varphi$ and $L(\varphi) = \int_{\Omega} f \varphi + \int_{\Gamma_N} g \varphi$.

Theorem 5.7 (Poincare inequality)

Let Ω be a domain with Lipschitz continuous boundary $\partial\Omega \subseteq \mathbb{R}^d$. Let $V = \{\varphi \in H^1(\Omega) : \varphi = 0 \text{ on } \Gamma_D\}$. If the $(d-1)$ -dimensional measure of Γ_D is positive ($|\Gamma_D| > 0$), there exists a constant $C_P > 0$ such that

$$\|u\|_{L^2(\Omega)} \leq C_P \|\nabla u\|_{L^2(\Omega)}, \quad \forall u \in V.$$



Fact: If Ω has a Lipschitz continuous boundary, $H^1(\Omega)$ is compactly embedded in $L^2(\Omega)$ (Rellich-Kondrachov Theorem).

Proof Assume the estimate does not hold. Then for any $n \in \mathbb{N}$, there exists $u_n \in V$ such that

$$\|u_n\|_{L^2(\Omega)} > n \|\nabla u_n\|_{L^2(\Omega)}.$$

Normalize the sequence by setting $\|u_n\|_{L^2(\Omega)} = 1$. Then we have:

$$\|\nabla u_n\|_{L^2(\Omega)} < \frac{1}{n}.$$

Thus, $\|u_n\|_{H^1(\Omega)}^2 = \|u_n\|_{L^2}^2 + \|\nabla u_n\|_{L^2}^2 \leq 1 + \frac{1}{n^2}$, so $\{u_n\}$ is bounded in $H^1(\Omega)$.

By the compact embedding result, there exists a subsequence (still denoted u_n) and a function $u \in L^2(\Omega)$ such that $u_n \rightarrow u$ strongly in $L^2(\Omega)$. Since $\|\nabla u_n\|_{L^2} \rightarrow 0$, $\{u_n\}$ is actually a Cauchy sequence in $H^1(\Omega)$ (because both L^2 parts and gradient parts converge).

Thus $u_n \rightarrow u$ strongly in $H^1(\Omega)$, which implies:

1. $\|\nabla u\|_{L^2} = \lim_{n \rightarrow \infty} \|\nabla u_n\|_{L^2} = 0 \implies u = \text{const.}$
2. Since $u_n \in V$, the trace $u|_{\Gamma_D} = 0$.

A constant function that is zero on a set of positive measure must be zero everywhere. Thus $u \equiv 0$. However, we assumed $\|u_n\|_{L^2} = 1$, so $\|u\|_{L^2} = \lim \|u_n\|_{L^2} = 1$. This contradicts $u \equiv 0$.

Remark The Poincare inequality implies that on the subspace V , the semi-norm $\|\nabla u\|_{L^2}$ is equivalent to the full H^1 norm. This ensures coercivity of $A(u, \varphi)$, allowing the application of the Lax-Milgram theorem.

Example 5.2

$$\begin{aligned} -\nabla \cdot (a \nabla u) &= f \quad \text{in } \Omega \\ u &= g \quad \text{on } \partial\Omega \end{aligned} \tag{5.8}$$

where $0 < a_0 \leq a(x) \leq a_1$.

Derivation: Multiply by test function φ (where $\varphi = 0$ on $\partial\Omega$):

$$-\int_{\Omega} \nabla \cdot (a \nabla u) \varphi = \int_{\Omega} a \nabla u \cdot \nabla \varphi - \int_{\partial\Omega} a(\nabla u \cdot \mathbf{n}) \underbrace{\varphi}_0 = \int_{\Omega} a \nabla u \cdot \nabla \varphi.$$

Define the affine space for the trial solution:

$$\tilde{V} = \{v \in H^1(\Omega) \mid v = g \text{ on } \partial\Omega\}.$$

Note: Usually we assume $g \in H^{1/2}(\partial\Omega)$ is the trace of some function $u_g \in H^1(\Omega)$.

Weak formulation: Find $u \in \tilde{V}$ satisfying

$$\int_{\Omega} a \nabla u \cdot \nabla \varphi = \int_{\Omega} f \varphi, \quad \forall \varphi \in H_0^1(\Omega).$$

Lifting (Homogenization): Let $u = w + u_g$, where $u_g \in H^1(\Omega)$ is a known extension of boundary data

g , and $w \in H_0^1(\Omega)$ is the unknown. Plugging this in:

$$\begin{aligned}\int_{\Omega} a \nabla(w + u_g) \cdot \nabla \varphi &= \int_{\Omega} f \varphi \\ \Rightarrow \int_{\Omega} a \nabla w \cdot \nabla \varphi &= \int_{\Omega} f \varphi - \int_{\Omega} a \nabla u_g \cdot \nabla \varphi = \langle \tilde{F}, \varphi \rangle.\end{aligned}$$

Now we can use Lax-Milgram on $H_0^1(\Omega)$ to find w , and then reconstruct $u = w + u_g$.

Chapter 6 Error of the Discrete System

6.1 Numerical Quadrature

We discuss the quadrature effects on the finite element approximation. Here we shall restrict our discussion to the two-dimensional space using triangular meshes.

6.1.1 Example Quadrature Schemes

Consider a triangle element τ with vertices V_1, V_2, V_3 and midpoints m_1, m_2, m_3 .

Several common quadrature rule (e.g., for the reference triangle) are:

$$\int_{\tau} f(x) dx \approx Q_{\tau} f = \frac{|\tau|}{3} \sum_{i=1}^3 f(v_i).$$

$$\int_{\tau} f(x) dx \approx \frac{|\tau|}{3} \sum_{i=1}^3 f(m_i).$$

$$\int_{\tau} f(x) dx \approx |\tau| \cdot f(b_{\tau}).$$

Definition 6.1

Q_{τ} is **exact** on $\mathcal{P}^k$ if:

$$Q_{\tau} \varphi = \int_{\tau} \varphi dx, \quad \forall \varphi \in \mathcal{P}^k$$



All the above schemes are exact for $\mathcal{P}^0$. What about $\mathcal{P}^1, \mathcal{P}^2$?

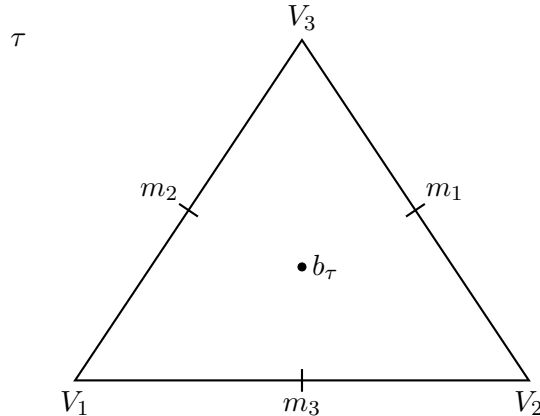
6.1.2 Discrete System Computation

In Finite Element Methods, we need to compute the stiffness matrix elements:

$$(A)_{ij} = \int_{\Omega} a(x) \nabla \varphi_j \cdot \nabla \varphi_i dx$$

We approximate it using quadrature:

$$(\tilde{A}_h)_{ij} = \sum_{\tau \in \mathcal{T}_h} Q_{\tau} (a(x) \nabla \varphi_j \cdot \nabla \varphi_i)$$



Similarly for the load vector:

$$F_j = \int_{\Omega} f(x) \varphi_j(x) dx \implies (\tilde{F}_h)_j = \sum_{\tau \in \mathcal{T}_h} Q_{\tau}(f(x) \varphi_j(x))$$

6.2 Weak Formulation with Quadrature

In general, we have the following weak formulation:

Find $u \in V$ such that:

$$A(u, \varphi) = \ell(\varphi), \quad \forall \varphi \in V \quad (6.1)$$

The standard Finite Element formulation reads:

Find $u_h \in V_h \subset V$ such that:

$$A(u_h, \varphi_h) = \ell(\varphi_h), \quad \forall \varphi_h \in V_h \quad (6.2)$$

But in practice, we cannot compute the integrals exactly. This leads to the **perturbed problem**:

Find $\tilde{u}_h \in V_h \subset V$ such that:

$$A_h(\tilde{u}_h, \varphi_h) = \ell_h(\varphi_h), \quad \forall \varphi_h \in V_h \quad (*) \quad (6.3)$$

For example:

$$\begin{aligned} A_h(w_h, \varphi) &= \sum_{\tau \in \mathcal{T}_h} Q_{\tau}(a(x) \nabla w_h \cdot \nabla \varphi) \\ \ell_h(\varphi) &= \sum_{\tau \in \mathcal{T}_h} Q_{\tau}(f \cdot \varphi) \end{aligned}$$

We consider two main issues:

1. The well-posedness of problem (*).
2. The error estimates between u and $\tilde{u}_h$.

Definition 6.2 (Uniform V_h -ellipticity)

The form $A_h(\cdot, \cdot)$ is uniformly V_h -elliptic if there exists $\alpha > 0$, independent of h , satisfying:

$$\alpha \|v\|_V^2 \leq A_h(v, v), \quad \forall v \in V_h \quad (6.4) \quad \clubsuit$$

Remark Condition (6.4) implies the existence and uniqueness of the discrete solution $\tilde{u}_h$ (Exercise).

6.3 Strang's First Lemma

Lemma 6.1 (Strang's First Lemma)

Assume that A_h is uniformly V_h -elliptic. Then there holds:

$$\|u - \tilde{u}_h\|_V \leq C \inf_{v_h \in V_h} \left\{ \|u - v_h\|_V + \sup_{w_h \in V_h} \frac{|A(v_h, w_h) - A_h(v_h, w_h)|}{\|w_h\|_V} \right\} + \sup_{w_h \in V_h} \frac{|l(w_h) - l_h(w_h)|}{\|w_h\|_V} \quad (6.5)$$

(Note: The constant C typically involves α and the continuity of A .)



Proof By the triangle inequality, we have for any $v_h \in V_h$:

$$\|u - \tilde{u}_h\|_V \leq \|u - v_h\|_V + \|v_h - \tilde{u}_h\|_V \quad (6.6)$$

Next, we estimate the second term $\|v_h - \tilde{u}_h\|_V$. Let $w_h = \tilde{u}_h - v_h$. Using the uniform ellipticity of A_h :

$$\alpha \|w_h\|_V^2 \leq A_h(w_h, w_h) = A_h(\tilde{u}_h - v_h, w_h) \quad (6.7)$$

Using the discrete equation $A_h(\tilde{u}_h, w_h) = l_h(w_h)$:

$$A_h(\tilde{u}_h - v_h, w_h) = l_h(w_h) - A_h(v_h, w_h) \quad (6.8)$$

We add and subtract terms to introduce the continuous forms $A(\cdot, \cdot)$ and $l(\cdot)$. Note that $A(u, w_h) = l(w_h)$ (exact solution consistency). Thus:

$$\begin{aligned} \alpha \|w_h\|_V^2 &\leq l_h(w_h) - A_h(v_h, w_h) \\ &= \underbrace{A(u, w_h) - A(v_h, w_h)}_{A(u-v_h, w_h)} + \underbrace{A(v_h, w_h) - A_h(v_h, w_h)}_{\text{Consistency Error in } A} + \underbrace{l_h(w_h) - l(w_h)}_{\text{Consistency Error in } l} \end{aligned}$$

Using the boundedness of A ($|A(u - v_h, w_h)| \leq \|A\| \|u - v_h\|_V \|w_h\|_V$):

$$\alpha \|w_h\|_V^2 \leq \|A\| \|u - v_h\|_V \|w_h\|_V + |A(v_h, w_h) - A_h(v_h, w_h)| + |l(w_h) - l_h(w_h)| \quad (6.9)$$

Dividing by $\|w_h\|_V$ (assuming $w_h \neq 0$):

$$\alpha \|\tilde{u}_h - v_h\|_V \leq \|A\| \|u - v_h\|_V + \frac{|A(v_h, w_h) - A_h(v_h, w_h)|}{\|w_h\|_V} + \frac{|l(w_h) - l_h(w_h)|}{\|w_h\|_V} \quad (6.10)$$

Combining this with the initial triangle inequality:

$$\|u - \tilde{u}_h\|_V \leq \left(1 + \frac{\|A\|}{\alpha}\right) \|u - v_h\|_V + \frac{1}{\alpha} \frac{|A(v_h, w_h) - A_h(v_h, w_h)|}{\|w_h\|_V} + \frac{1}{\alpha} \frac{|l(w_h) - l_h(w_h)|}{\|w_h\|_V} \quad (6.11)$$

Taking the infimum over $v_h \in V_h$ and the supremum over $w_h \in V_h$ yields the lemma.

6.4 Linear Iterative Methods

Let us consider solving the linear system:

$$Ax = f \quad (6.12)$$

where A is a symmetric positive-definite (SPD) matrix.

6.4.1 The Picard Iteration

We start with the simplest iterative method for solving (6.12). Given an initial iterate x^0 , we set:

$$x^{n+1} = x^n + \tau(f - Ax^n), \quad n = 0, 1, \dots \quad (6.13)$$

Let the error be defined as $e^n = x - x^n$. Then it is immediate that:

$$e^{n+1} = (I - \tau A)e^n.$$

Let $\|I - \tau A\|$ denote the operator norm induced by the Euclidean norm $\|\cdot\|_2$, namely:

$$\|I - \tau A\| = \sup_{v \in \mathbb{R}^N} \frac{\|(I - \tau A)v\|}{\|v\|} = \sup_{\lambda \in \sigma(A)} |1 - \tau \lambda|.$$

Here $\sigma(A)$ denotes the spectrum of A , and λ_1, λ_N are respectively the smallest and largest eigenvalues of A . It follows that:

$$\|e^n\| \leq \|I - \tau A\| \|e^{n-1}\| \leq \dots \leq \|I - \tau A\|^n \|e^0\|.$$

The convergence rate is determined by:

$$\|I - \tau A\| = 1 - \frac{\lambda_1}{\lambda_N} = 1 - \frac{1}{\kappa(A)}, \quad (6.14)$$

where $\kappa(A) = \frac{\lambda_N}{\lambda_1}$ is known as the spectral condition number of A .

Clearly, the bound is minimized when we choose $\tau = \frac{2}{\lambda_1 + \lambda_N}$, which gives:

$$\|I - \tau A\| = \frac{\lambda_N - \lambda_1}{\lambda_N + \lambda_1} = \frac{\kappa(A) - 1}{\kappa(A) + 1}. \quad (6.15)$$

Note that this is not much of an improvement over the simple choice of $\tau = \lambda_N^{-1}$ when $\kappa(A)$ is large.

6.4.2 Preconditioning

From the above discussion, it is clear that even with the best choice of τ , the scheme will converge slowly if the condition number of A is large.

The idea of preconditioning is to apply a "preconditioner" to the system and then iterate. The rate of convergence can be improved by the use of an appropriately defined preconditioner.

More precisely, let B be another symmetric and positive definite operator. Consider applying the iteration to the preconditioned equation:

$$BAx = Bf.$$

The iteration becomes:

$$x^{n+1} = x^n + \tau B(f - Ax^n). \quad (6.16)$$

Let $e^n = x - x^n$. Then $e^n = (I - \tau BA)e^{n-1}$. We analyze this using the A -inner product:

$$(u, v)_A = (Au, v), \quad \forall u, v \in \mathbb{R}^N.$$

The associated norm is $\|\cdot\|_A = (\cdot, \cdot)_A^{1/2}$. Note that BA is symmetric with respect to $(\cdot, \cdot)_A$ and is positive definite. The analysis is syntactically similar to the previous discussion but using the norm $\|\cdot\|_A$:

$$\|I - \tau BA\|_A = \sup_{v \in \mathbb{R}^N} \frac{\|(I - \tau BA)v\|_A}{\|v\|_A} = \sup_{\hat{\lambda} \in [\hat{\lambda}_1, \hat{\lambda}_N]} |1 - \tau \hat{\lambda}|,$$

where $\hat{\lambda}_1, \hat{\lambda}_N$ are respectively the smallest and largest eigenvalues of BA . The convergence rate can be estimated in terms of $\kappa(BA)$ as before.

If τ is chosen optimally, the spectral radius is $\rho(I - \tau BA) = \max_i |1 - \tau \hat{\lambda}_i|$. The iteration converges if and only if $|1 - \tau \hat{\lambda}_i| < 1$, which implies $0 < \tau \hat{\lambda}_i < 2$.

6.5 The Conjugate Gradient Method

6.5.1 Krylov Space and Minimization Problem

We define the Krylov space $\mathcal{K}_n(A, r^0) = \text{span}\{r^0, Ar^0, \dots, A^{n-1}r^0\}$. Consider the minimization problem:

$$J(x) = \min_{y \in \mathbb{R}^N} J(y),$$

where the functional is defined as:

$$J(y) = \frac{1}{2}(Ay, y) - (f, y) = \frac{1}{2}(A(x - y), (x - y)) - \frac{1}{2}(Ax, x).$$

Successive approximations to the solution can be constructed by minimizing $J(y)$ over the Krylov space. Given x^0 , let $r^0 = f - Ax^0$. Then $x^n = x^0 + \delta^n$ with $\delta^n \in \mathcal{K}_n$ being the minimizer of $J(y)$ over the affine space $\{x^0\} + \mathcal{K}_n$.

6.5.2 Algorithms

6.5.2.1 Algorithm (Standard CG)

Given x^0 arbitrary. Set $r^0 = f - Ax^0$ and $p^0 = r^0$. For $n = 0, 1, \dots$ until convergence, compute:

1. $x^{n+1} = x^n + \alpha_n p^n$, where $\alpha_n = \frac{(r^n, p^n)}{(Ap^n, p^n)}$
2. $r^{n+1} = r^n - \alpha_n Ap^n$
3. $p^{n+1} = r^{n+1} + \beta_n p^n$, where $\beta_n = -\frac{(Ap^n, r^{n+1})}{(Ap^n, p^n)}$

6.5.2.2 Algorithm (PCG)

Given x^0 arbitrary. Set $r^0 = f - Ax^0$ and $z^0 = Br^0$, $p^0 = z^0$. For $n = 0, 1, \dots$ until convergence, compute:

1. $x^{n+1} = x^n + \alpha_n p^n$, where $\alpha_n = \frac{(r^n, z^n)}{(Ap^n, p^n)}$
2. $r^{n+1} = r^n - \alpha_n Ap^n$
3. $z^{n+1} = Br^{n+1}$
4. $p^{n+1} = z^{n+1} + \beta_n p^n$, where $\beta_n = \frac{(r^{n+1}, z^{n+1})}{(r^n, z^n)}$

6.6 Convergence Analysis

Theorem: Let $e^n = x - x^n$. For CG:

$$\|e^n\|_A \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^n \|e^0\|_A, \quad \text{where } \kappa = \kappa(A). \quad (6.17)$$

For PCG:

$$\|e^n\|_A \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^n \|e^0\|_A, \quad \text{where } \kappa = \kappa(BA). \quad (6.18)$$

Remark: Note that $\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1} < \frac{\kappa-1}{\kappa+1}$, so CG converges faster than the simple iterative method.

6.7 Model Problem: Finite Element Application

Consider the problem:

$$\begin{cases} -\Delta u = f & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases} \quad (6.19)$$

The weak formulation is: Find $u \in H_0^1(\Omega)$ such that:

$$\int_{\Omega} \nabla u \cdot \nabla v = \int_{\Omega} f v, \quad \forall v \in H_0^1(\Omega).$$

6.7.1 Multilevel Finite Element Constructions

Let $\{\mathcal{T}_k\}_{k=1}^\infty$ be nested triangulations obtained by subdividing each triangle into four pieces by connecting midpoints of edges. Let V_k be the continuous linear Finite Element space of $\mathcal{T}_k$. So,

$$V_1 \subset V_2 \subset V_3 \subset \cdots \subset V_J \subset H_0^1(\Omega).$$

Let h_k be the mesh size of $\mathcal{T}_k$. We denote the corresponding linear system $A_k U = F$, where:

$$(A_k)_{ij} = \int_{\Omega} \nabla \phi_j \cdot \nabla \phi_i, \quad F_j = \int_{\Omega} f \phi_j.$$

Here $U = (u_1, \dots, u_N)$ so that $u_h = \sum_{i=1}^N u_i \phi_i$.

For each $T \in \mathcal{T}_k$, we note that for v_h linear in T :

$$\|v_h\|_{L^2(T)}^2 \approx h_k^2 \sum_{i=1}^3 v_h^2(v_i),$$

where v_i are the vertices of T . Summing over triangles:

$$\|v_h\|_{L^2(\Omega)}^2 \sim h^2 \sum_{i=1}^N v_i^2.$$

By Poincaré inequality:

$$h^2 \sum_{i=1}^N v_i^2 \leq C \|v_h\|_{L^2(\Omega)}^2 \leq C \int_{\Omega} |\nabla v_h|^2 = C V^T A_k V.$$

Thus, $V^T A_k V \geq C h_k^2 |V|^2$. Since A_k is SPD, this means A_k 's smallest eigenvalue is bounded below by $C h^2$.

On the other hand, it can be shown that the largest eigenvalue of A_k is uniformly bounded above. Thus, the condition number behaves as:

$$\kappa(A_k) = \frac{\lambda_N}{\lambda_1} \leq C h_k^{-2}. \quad (6.20)$$

One can also show that $\kappa(A_k) \geq c h_k^{-2}$ (Exercise). This means the problem is rather ill-conditioned, and iterative methods will converge slowly.